EMV® Terminal Type Approval Bulletin No. 242
Second Edition, January 2021

Terminal Level 2 Test Cases
v4.3i Update

Applicability

This Bulletin applies to:

- *EMV® Terminal Type Approval - Level 2 Test Cases – version 4.3i*

Related Documents

This Bulletin should be read in conjunction with:

- *EMV® Terminal Type Approval - Level 2 Test Cases – version 4.3i*

Effective Date

- January 31th 2021

Description

Based on EMVCo's continued review of the *EMVCo Type Approval Terminal Level 2 Test Cases v4.3i,* the following updates are necessary, and shall be applied to any type approval session beginning on, or after, the effective date cited above.

Update to EMV® Terminal Type Approval - Level 2 Test Cases – version 4.3i

Section 5.5 – Default Values Setting Updates

Introduction

A Bitmasks checking mechanism at the test tool level shall be implemented to verify that the TVR and the TSI bits, which are not used depending on the test cases conditions, stay at '0' and are not wrongly set to '1'.
As a result of this bitmasks checking, a new pass criteria is added in all test cases, to reflect it.

Three levels of bitmask are defined to create the 'Un-set TVR bits' (and TSI) Pass Criteria:

- Terminal configuration setting bitmask (example: if CDA is not supported, TVR bit CDA failed can never be set), named ICS-bitmask
- LT (Card) setting bitmask (example if AIP of LT does not support DDA, TVR bit DDA failed can never be set), named LT-bitmask
- Transaction bitmask (example if Transaction date is earlier then Expiration, TVR bit Expired can never be set), named Transaction-bitmask

Note that even if the Transaction-bitmask is defined here, it will be not used for the time being.

TVR ICS-bitmask

The ICS-bitmask value is defined by the tool at the start of a kernel configuration test session. This bitmask is depending on the supported options of this kernel configuration (listed in the ICS) and stay at the same value during the whole kernel configuration testing. It must be recalculated based on the supported options for each kernel configuration under test.

The ICS-bitmask is limited to some TVR bits and are depending on the related ICS question(s) other TVR bit shall have the value ‘0’ in the bitmask:

TVR	TVR Name	Related ICS question of the configuration	bitmask has the value:
B1b8	Offline data authentication was not performed		0
B1b7	SDA failed	SDA and DDA = NO	1
B1b6	ICC data missing		0
B1b5	Card appears on terminal exception file	Exception File = NO	1
B1b4	DDA failed	SDA and DDA = NO	1
B1b3	CDA failed	CDA = NO	1
B1b2	SDA Selected	SDA and DDA = NO	1
B1b1	RFU	Always set	1
B2b8	ICC and terminal have different application versions		0
B2b7	Expired application		0
B2b6	Application not yet effective		0
B2b5	Requested service not allowed for card product		0
B2b4	New card	Velocity Checking = NO	1
B2b3	RFU	Always set	1
B2b2	RFU or Biometric not performed or not successful	Biometric = No	1
B2b1	RFU or Biometric template format supported	Biometric = No	1
B3b8	Cardholder verification was not successful		0
B3b7	Unrecognised CVM		0
B3b6	PIN Try Limit exceeded	Offline Enciphered PIN AND Plaintext PIN = NO	1
B3b5	PIN entry required and PIN pad not present or not working	Offline Enciphered PIN AND Plaintext PIN AND Online Enciphered PIN = NO	1
B3b4	PIN entry required, PIN pad present, but PIN was not entered	Offline Enciphered PIN AND Plaintext PIN AND Online Enciphered PIN = NO	1
B3b3	Online PIN entered	Online Enciphered PIN = NO	1
B3b2	RFU	Always set	1
B3b1	RFU	Always set	1
B4b8	Transaction exceeds floor limit	Floor Limit Checking = NO	1
B4b7	Lower consecutive offline limit exceeded	Velocity Checking = NO	1
B4b6	Upper consecutive offline limit exceeded	Velocity Checking = NO	1
B4b5	Transaction selected randomly for online processing	Random Transaction Selection = NO	1
B4b4	Merchant forced transaction online		0
B4b3	RFU or Biometric Try Limit not exceeded	Biometric = No	1
B4b2	RFU or A selected Biometric Type not supported	Biometric = No	1
B4b1	RFU	Always set	1
B5b8	Default TDOL used	Default TDOL = NO	1
B5b7	Issuer authentication failed		0
B5b6	Script processing failed before final GENERATE AC		0
B5b5	Script processing failed after final GENERATE AC		0
B5b4	RFU	Always set	1

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B5b3	RFU	Always set	1
B5b2	RFU	Always set	1
B5b1	RFU	Always set	1

Note: even if not related to ICS-bitmask, the un-setting of RFU TVR bits is checked here.

TVR LT-bitmask

The LT-bitmask value is defined by the tool for a kernel configuration testing. This bitmask is depending on the LT setting used for a batch of test cases. LT-bitmask must be updated anytime the LT is changed:

- AIP
- AUC
- TDOL presence
- ATC value = 0
- Effective Date present
- Last Online ATC absent
- Lower Consecutive Limit and Upper Consecutive Limit absent
- …

The LT-bitmask is limited to the following TVR bits (any other TVR bit shall have the value '0' in this bitmask):

TVR	TVR Name	LT Data setting	bitmask has the value:
B1b8	Offline data authentication was not performed		0
B1b7	SDA failed	AIP States SDA is not supported	1
B1b6	ICC data missing	All Mandatory Data present in LT	1
B1b5	Card appears on terminal exception file		0
B1b4	DDA failed	AIP States DDA is not supported	1
B1b3	CDA failed	AIP States CDA is not supported	1
B1b2	SDA Selected	AIP States SDA is not supported	1
B1b1	RFU		0
B2b8	ICC and terminal have different application versions	ICC application version not present in LT	1
B2b7	Expired application		0
B2b6	Application not yet effective	Effective Data not present in LT	1
B2b5	Requested service not allowed for card product	(AUC not present in LT) OR (AUC value = 'FFC0')	1
B2b4	New card	(ATC different from '0') OR (LOATC # '0') OR (LCOL AND UCOL not present)	1
B2b3	RFU		0
B2b2	RFU or Biometric not performed or not successful		0
B2b1	RFU or Biometric template format supported		0
B3b8	Cardholder verification was not successful	AIP states Cardholder Verification is not supported	1
B3b7	Unrecognised CVM	(AIP states Cardholder Verification is not supported) OR (CVM list uses only EMVCo recognized Code)	1
B3b6	PIN Try Limit exceeded	(AIP states Cardholder Verification is not supported) OR (CVM list does not contain Offline Enciphered PIN AND Plaintext PIN)	1

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B3b5	PIN entry required and PIN pad not present or not working	(AIP states Cardholder Verification is not supported) OR (CVM list does not contain any PIN)	1
B3b4	PIN entry required, PIN pad present, but PIN was not entered	(AIP states Cardholder Verification is not supported) OR (CVM list does not contain any PIN)	1
B3b3	Online PIN entered	(AIP states Cardholder Verification is not supported) OR (CVM List does not contain Online PIN)	1
B3b2	RFU		0
B3b1	RFU		0
B4b8	Transaction exceeds floor limit		0
B4b7	Lower consecutive offline limit exceeded	Lower consecutive offline limit not present in LT	1
B4b6	Upper consecutive offline limit exceeded	Upper consecutive offline limit not present in LT	1
B4b5	Transaction selected randomly for online processing		0
B4b4	Merchant forced transaction online		0
B4b3	RFU or Biometric Try Limit not exceeded		0
B4b2	RFU or A selected Biometric Type not supported		0
B4b1	RFU		0
B5b8	Default TDOL used	TDOL present in LT	1
B5b7	Issuer authentication failed		0
B5b6	Script processing failed before final GENERATE AC		0
B5b5	Script processing failed after final GENERATE AC		0
B5b4	RFU		0
B5b3	RFU		0
B5b2	RFU		0
B5b1	RFU		0

TVR Transaction-bitmask

All bits of the Transaction-bitmask shall be set to zero for the time being.

TVR	TVR Name	LT Data setting	bitmask has the value:
B1b8	Offline data authentication was not performed		0
B1b7	SDA failed		0
B1b6	ICC data missing		0
B1b5	Card appears on terminal exception file		0
B1b4	DDA failed		0
B1b3	CDA failed		0
B1b2	SDA Selected		0
B1b1	RFU		0
B2b8	ICC and terminal have different application versions		0
B2b7	Expired application		0
B2b6	Application not yet effective		0
B2b5	Requested service not allowed for card product		0

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B2b4	New card		0
B2b3	RFU		0
B2b2	RFU or Biometric not performed or not successful		0
B2b1	RFU or Biometric template format supported		0
B3b8	Cardholder verification was not successful		0
B3b7	Unrecognised CVM		0
B3b6	PIN Try Limit exceeded		0
B3b5	PIN entry required and PIN pad not present or not working		0
B3b4	PIN entry required, PIN pad present, but PIN was not entered		0
B3b3	Online PIN entered		0
B3b2	RFU		0
B3b1	RFU		0
B4b8	Transaction exceeds floor limit		0
B4b7	Lower consecutive offline limit exceeded		0
B4b6	Upper consecutive offline limit exceeded		0
B4b5	Transaction selected randomly for online processing		0
B4b4	Merchant forced transaction online		0
B4b3	RFU or Biometric Try Limit not exceeded		0
B4b2	RFU or A selected Biometric Type not supported		0
B4b1	RFU		0
B5b8	Default TDOL used		0
B5b7	Issuer authentication failed		0
B5b6	Script processing failed before final GENERATE AC		0
B5b5	Script processing failed after final GENERATE AC		0
B5b4	RFU		0
B5b3	RFU		0
B5b2	RFU		0
B5b1	RFU		0

Bitmask Checking Mechanism

The Tool shall implement, for each transaction run in all test cases, the following checking function (applicable for all TVR bits, one by one) based on the previous bitmasks:

IF (TVR ByteXbitY is set to ‘1’) AND ((ICS-bitmask ByteXbitY = ‘1) OR (LT-bitmask ByteXbitY = ‘1) OR (Transaction-bitmask ByteXbitY = ‘1)) # ZERO THEN Pass Criteria = error on ByteXbitY

‘Un-set TVR bits’ Pass Criteria

The following Pass Criteria apply to all test cases individually:

TVR bits, where ICS-bitmask and/or the LT-bitmask equivalent bits are set to ‘1’, shall be set to '0' received at 1ST GENERATE AC.

TSI-ICS-bitmask

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The ICS-bitmask value is defined by the tool at the start of a kernel configuration test session. This bitmask is depending on the supported options of this kernel configuration (listed in the ICS) and stay at the same value during the whole kernel configuration testing. It must be recalculated based on the supported options for each kernel configuration under test.
The ICS-bitmask is limited to some TSI bits and are depending on the related ICS question(s) other TSI bit shall have the value '0' in the bitmask:

TSI	TSI Name	Related ICS question of the configuration	bitmask has the value:
B1b8	Offline data authentication was performed	(SDA and DDA) AND CDA = NO	1
B1b7	Cardholder verification authentication was performed		0
B1b6	Card risk management was performed		0
B1b5	Issuer authentication was performed	Offline Only = YES	1
B1b4	Terminal risk management was performed	(Velocity Checking) AND (Random Transaction Selection) AND (Floor Limit) = NO	1
B1b3	Script processing was performed	Offline Only = YES	1
B1b2	RFU		1
B1b1	RFU		1
B2b8	RFU		1
B2b7	RFU		1
B2b6	RFU		1
B2b5	RFU		1
B2b4	RFU		1
B2b3	RFU		1
B2b2	RFU		1
B2b1	RFU		1

Note: even if not related to ICS-bitmask, the un-setting of RFU TSI bits is checked here.

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TSI-LT-bitmask

The LT-bitmask value is defined by the tool for a kernel configuration testing. This bitmask is depending on the LT setting used for a batch of test cases. LT-bitmask must be updated anytime the LT is changed:

- AIP
- …

The LT-bitmask is limited to the following TSI bits (any other TSI bit shall have the value '0' in this bitmask):

TSI	TSI Name	LT Data setting	bitmask has the value:
B1b8	Offline data authentication was performed	AIP states SDA AND DDA AND CDA are not supported	1
B1b7	Cardholder verification authentication was performed	AIP states Cardholder Verification is not supported OR CVM List is not present in LT	1
B1b6	Card risk management was performed		0
B1b5	Issuer authentication was performed		0
B1b4	Terminal risk management was performed		0
B1b3	Script processing was performed		0
B1b2	RFU		0
B1b1	RFU		0
B2b8	RFU		0
B2b7	RFU		0
B2b6	RFU		0
B2b5	RFU		0
B2b4	RFU		0
B2b3	RFU		0
B2b2	RFU		0
B2b1	RFU		0

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TSI-Transaction-bitmask

All bits of the TSI-Transaction-bitmask shall be set to zero for the time being.

TSI	TSI Name	LT Data setting	bitmask has the value:
B1b8	Offline data authentication was performed		0
B1b7	Cardholder verification authentication was performed		0
B1b6	Card risk management was performed		0
B1b5	Issuer authentication was performed		0
B1b4	Terminal risk management was performed		0
B1b3	Script processing was performed		0
B1b2	RFU		0
B1b1	RFU		0
B2b8	RFU		0
B2b7	RFU		0
B2b6	RFU		0
B2b5	RFU		0
B2b4	RFU		0
B2b3	RFU		0
B2b2	RFU		0
B2b1	RFU		0

Bitmask Checking Mechanism

The Tool shall implement, for each transaction run in all test cases, the following checking function (applicable for all TSI bits, one by one) based on the previous bitmasks:

IF (TSI ByteXbitY is set to ‘1’) AND ((ICS-bitmask ByteXbitY = ‘1) OR (LT-bitmask ByteXbitY = ‘1) OR (Transaction-bitmask ByteXbitY = ‘1)) # ZERO THEN Pass Criteria = error on ByteXbitY

‘Un-set TSI bits’ Pass Criteria

The following Pass Criteria apply to all test cases individually:

TSI bits, where TSI-ICS-bitmask and/or the TSI-LT-bitmask equivalent bits are set to ‘1’, shall be set to '0' received at 1ST GENERATE AC.

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Section 7 – Test Cases Updates

New test cases include:

2CA.122.06	2CA.122.15	2CA.122.20	2CA.122.25	2CM.083.03
2CA.122.11	2CA.122.16	2CA.122.21	2CB.031.10	
2CA.122.12	2CA.122.17	2CA.122.22	2CC.146.00	
2CA.122.13	2CA.122.18	2CA.122.23	2CM.083.01	
2CA.122.14	2CA.122.19	2CA.122.24	2CM.083.02	

Revised test cases include:

2CA.015.00	2CC.071.00	2CJ.002.10	2CJ.047.00	2CJ.060.00
2CA.015.01	2CI.021.00	2CJ.002.11	2CJ.048.00	2CJ.062.00
2CA.015.02	2CJ.002.00	2CJ.002.12	2CJ.049.00	2CK.005.01
2CA.015.03	2CJ.002.01	2CJ.002.13	2CJ.050.00	2CM.063.00
2CA.124.00	2CJ.002.02	2CJ.002.14	2CJ.051.00	2CM.064.00
2CA.124.01	2CJ.002.03	2CJ.035.00	2CJ.052.00	2CM.067.00
2CA.124.02	2CJ.002.04	2CJ.041.00	2CJ.053.00	2CM.083.00
2CA.124.03	2CJ.002.05	2CJ.042.00	2CJ.054.00	
2CA.130.00	2CJ.002.06	2CJ.043.00	2CJ.056.00	
2CB.024.00	2CJ.002.07	2CJ.044.00	2CJ.057.00	
2CB.031.00	2CJ.002.08	2CJ.045.00	2CJ.058.00	
2CB.031.09	2CJ.002.09	2CJ.046.00	2CJ.059.00	

Deleted test cases include:

2CA.054.01

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2CA.015.00 Processing DOLs: data absent from ICC

Test No: **2CA.015.00** - *Revision* 43j - *Number of sub cases:* 6

Objective: To verify that whenever a data object is in the list and is meaningful to the IUT but represents optional static data absent from the ICC, the portion of the command field corresponding is filled with hexadecimal zeroes

Reference EMV 4.3: Book 3 - Section 5.4 - Rules for Using a Data Object List (DOL)

Terminal Configuration: NA

Terminal Configuration (New format): [NA]

Conditions:

- Case 01: In addition to the *default values* the CDOL1 of LT contains an optional ICC static data that is absent from the LT (for example the Issuer Action Code – Default, tag '9F0D'), this data is located in the middle of the CDOL1.
- Case 02: In addition to the *default values* the CDOL2 of LT contains an optional ICC static data that is absent from the LT and LT request ARQC on first GENERATE AC (for example the Issuer Action Code – Default, tag '9F0D'), this data is located in the middle of the CDOL2.
- Case 03: CDOL1 request the TC Hash value. The TDOL of LT contains (in this order) the Acquirer Identifier '9F01' (with a non-zero value), an optional ICC static data that is absent from the LT (for example the Issuer Action Code – Default, tag '9F0D') and the Merchant Identifier '9F16' (with a non-zero value)
- Case 04: In addition to the *default values* the CDOL1 of LT contains an optional ICC static data that is absent from the LT (for example the Issuer Action Code – Default, tag '9F0D'), this data is the last in the CDOL.
- Case 05: In addition to the *default values* the CDOL2 of LT contains an optional ICC static data that is absent from the LT and LT request ARQC on first GENERATE AC (for example the Issuer Action Code – Default, tag '9F0D'), this data is the last in the CDOL.
- Case 06: CDOL1 request the TC Hash value. The TDOL of LT contains (in this order) the Acquirer Identifier '9F01', the Merchant Identifier '9F16' (both with a non-zero value) and an optional ICC static data that is absent from the LT (for example the Issuer Action Code – Default, tag '9F0D').

Action: NA

Procedure: Application in LT is selected and transaction is performed with LT (in particular the DOL processing).

Pass Criteria:

- The LT shall receive the DOL with portion of the DOL field representing the Data Object filled with hexadecimal zeroes (portion has the same length as the Data Object in DOL).

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2CA.015.00 Processing DOLs: data absent from ICC

- Case 03 & 06: TC hash value received a First Generate AC shall be have the same value as the one computed (with ‘00’ value for the data absent)

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2CA.015.01 Processing DOLs: data absent from ICC (2)

Test No: **2CA.015.01** - *Revision* 43j - *Number of sub cases:* 2

Objective: To verify that whenever a data object is in the list and is meaningful to the IUT but represents optional static data absent from the ICC, the portion of the command field corresponding is filled with hexadecimal zeroes.
By using Dynamic Data Authentication.

Reference EMV 4.3: Book 3 - Section 5.4 - Rules for Using a Data Object List (DOL)

Terminal Configuration: [DDA] supported

Terminal Configuration (New format): [DDA]

Conditions:

- Case 01: The DDOL of LT requests the Unpredictable Number ‘9F37’ and an optional ICC static data that is absent from the LT (for example the Issuer Action Code – Default, tag ‘9F0D’)
- Case 02: The DDOL of LT requests an optional ICC static data that is absent from the LT (for example the Issuer Action Code – Default, tag ‘9F0D’) and the Unpredictable Number ‘9F37’

Action: NA

Procedure: Application in LT is selected and transaction is performed with LT (in particular the DOL processing).

Pass Criteria: The terminal shall return an INTERNAL AUTH command to the LT with portion of the DOL field representing the Data Object filled with hexadecimal zeroes (portion has the same length as the Data Object in DOL).

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2CA.015.02 Processing DOLs: data absent from Terminal

Test No: **2CA.015.02** - *Revision* 43j - *Number of sub cases:* 8

Objective: To verify that whenever a data object is in the list and is meaningful to the IUT but represents optional static data absent from the terminal, the portion of the command field corresponding is filled with hexadecimal zeroes

Reference EMV 4.3: Book 3 - Section 5.4 - Rules for Using a Data Object List (DOL)

Terminal Configuration:

Terminal Configuration (New format): [TRMDabsent]

Conditions:

- Case 01: In addition to the *default values* the CDOL1 of LT contains an optional terminal data that is absent: Terminal Risk Management Data ‘9F1D’, this data is located in the middle of the CDOL1.
- Case 02: In addition to the *default values* the CDOL2 of LT contains an optional terminal data that is absent: Terminal Risk Management Data ‘9F1D’, this data is located in the middle of the CDOL2.
- Case 03: CDOL1 request the TC Hash value. The TDOL of LT contains (in this order) the Acquirer Identifier ‘9F01’ (with a non-zero value), an optional terminal data that is absent: Terminal Risk Management Data ‘9F1D’ and the Merchant Identifier ‘9F16’ (with a non-zero value)
- Case 04: In addition to the *default values* the CDOL1 of LT contains an optional terminal data that is absent: Terminal Risk Management Data ‘9F1D’, this data is the last in the CDOL.
- Case 05: In addition to the *default values* the CDOL2 of LT contains an optional terminal data that is absent: Terminal Risk Management Data ‘9F1D’, this data is the last in the CDOL.
- Case 06: CDOL1 request the TC Hash value. The TDOL of LT contains (in this order) the Acquirer Identifier ‘9F01’, the Merchant Identifier ‘9F16’ (both with a non-zero value) and an optional terminal data that is absent: Terminal Risk Management Data ‘9F1D’.
- Case 07: The PDOL of LT contains the same data as the *default values* of CDOL2 and an optional terminal data that is absent and located in the middle of the PDOL: Terminal Risk Management Data ‘9F1D’.
- Case 08: The PDOL of LT contains the same data as the *default values* of CDOL2 and an optional terminal data that is absent and located at the end of the PDOL: Terminal Risk Management Data ‘9F1D’.

Action: NA

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2CA.015.02 Processing DOLs: data absent from Terminal

Procedure: Application in LT is selected and transaction is performed with LT (in particular the DOL processing).

Pass Criteria:

- The LT shall receive the DOL with portion of the DOL field representing the Data Object filled with hexadecimal zeroes (portion has the same length as the Data Object in DOL).
- Case 03 & 06: TC hash value received a First Generate AC shall have the same value as the one computed (with '00' value for the data absent)

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2CA.015.03 Processing DOLs: data absent from Terminal (2)

Test No: **2CA.015.03** - *Revision* 43j - *Number of sub cases:* 2

Objective: To verify that whenever a data object is in the list and is meaningful to the IUT but represents optional static data absent from the ICC, the portion of the command field corresponding is filled with hexadecimal zeroes.
By using Dynamic Data Authentication.

Reference EMV 4.3: Book 3 - Section 5.4 - Rules for Using a Data Object List (DOL)

Terminal Configuration:

Terminal Configuration (New format): [DDA] AND [TRMDabsent]

Conditions:

- Case 01: The DDOL of LT requests the Unpredictable Number ‘9F37’ and contain an optional terminal data that is absent: Terminal Risk Management Data ‘9F1D’,
- Case 02: The DDOL of LT requests an optional terminal data that is absent: Terminal Risk Management Data ‘9F1D’ and the Unpredictable Number ‘9F37’.

Action: NA

Procedure: Application in LT is selected and transaction is performed with LT (in particular the DOL processing).

Pass Criteria: The terminal shall return an INTERNAL AUTH command to the LT with portion of the DOL field representing the Data Object filled with hexadecimal zeroes (portion has the same length as the Data Object in DOL).

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2CA.122.06 EMV reserved Data objects processing (4)

Test No: **2CA.122.06** - *Revision* 43j - *Number of sub cases:* 3

Objective: To ensure that the terminal ignores EMV reserved Data Objects (not used by EMV).

Reference EMV 4.3: Book 3 – Annex B1 Coding of the Tag Field of BER-TLV Data Objects

Book Bulletins - Section SB 185 - Bulletin n°185, Biometric Terminal Specification (March 2017)

Terminal Configuration: [Biometric] not supported AND [PSE] supported

Terminal Configuration (New format): NOT[Biometric] AND [PSE]

Conditions:

- LT AID value is B000001234
- Terminal AID Value is B000001234
- PSE contains a directory file with entries for ADF.
- PDOL, CDOL1, CDOL2 and TDOL of LT contains EMV primitive tags reserved but not used in the EMV specification: ‘9F30’ and ‘9F31’
- Case 01: FCI Issuer discretionary Data (tag BF0C) of PSE contains the following EMV primitive tags reserved but not used in the EMV specification: ‘9F30’ and ‘9F31 with values different from zero.
- Case 02: FCI (tag 6F) of PSE contains the following EMV primitive tags reserved but not used in the EMV specification: ‘9F30’ and ‘9F31’ with values different from zero.
- Case 03: Template '73' of ADF Directory Entry contains the following EMV primitive tags reserved but not used in the EMV specification: ‘9F30’ and ‘9F31’ with values different from zero.

Action: NA

Procedure: Application in LT is selected and transaction is processed with LT until approval.

Pass Criteria:

- The terminal shall process the transaction until completion, by requesting a TC.
- Values of the reserved tags '9F30' and '9F31' received by the LT via the DOLs are set to “00’ (Note: for CDOL2 this is applicable only if a 2nd GENERATE AC has been performed)

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2CA.122.11 EMV reserved Data objects processing (8)

Test No: **2CA.122.11** – *Revision* 43j – *Number of sub cases:* 5

Objective: To ensure that the terminal ignores EMV reserved Data Objects (not used by EMV).

Reference EMV 4.3: Book 3 – Annex B1 Coding of the Tag Field of BER-TLV Data Objects

Book Bulletins - Section SB 197 - Bulletin n°197, Tokenisation Data Objects – Token Requestor ID and Last 4 Digits of PAN (May 2017)

Book Bulletins - Section SB 185 - Bulletin n°185, Biometric Terminal Specification (March 2017)

Book Bulletins - Section SB 231 - Bulletin n°231, Issuer Identification Number Extended (IINE) (December 2020)

Terminal Configuration:

Terminal Configuration (New format): [NA]

Conditions:

- LT AID value is A0000000031010
- Terminal AID Value is VISA A0000000031010
- PDOL, CDOL1, CDOL2 and TDOL of LT contains all EMV primitive tags reserved but not used in the EMV specification, (for example ‘9F28’and ‘9F29’, …) in the range ‘80’ to ‘9E’ and ‘9F01’ to ‘9F4F’.
- All Data Elements listed in the Specification Bulletin SB 185, Table 34 are excluded from the below lists.
- All Data Elements listed in the Specification Bulletin SB 197, Table 34 are excluded from the below lists.
- All Data Elements listed in the Specification Bulletin SB 231, Table 34 are excluded from the below lists.
- Case 01: FCI (tag 6F) of ADF contains additionally all EMV primitive tags reserved but not used in the EMV specification, (for example ‘9F28’and ‘9F29’, …) in the range ‘80’ to ‘9E’ and ‘9F01’ to ‘9F4F’ with values different from zero within the FCI template
- Case 02: FCI Issuer discretionary Data (tag BF0C) of ADF contains additionally all EMV primitive tags reserved but not used in the EMV specification, (for example ‘9F0C’, ‘9F28’and ‘9F29’, …) in the range ‘80’ to ‘9E’ and ‘9F01’ to ‘9F4F’ with values different from zero
- Case 03: Response to GENERATE AC contains all EMV primitive tags reserved but not used in the EMV specification, (for example ‘9F28’and ‘9F29’, …) in the range ‘80’ to ‘9E’ and ‘9F01’ to ‘9F4F’ with values different from zero and shall be encoded in template 77
- Case 04: Records contains: all EMV primitive tags reserved but not used in the EMV specification, (for example ‘9F28’and ‘9F29’, …) in

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the range ‘80’ to ‘9E’ and ‘9F01’ to ‘9F4F’ with values different from zero (splitted over SFI 1 to 10).

- Case 05: Response to GET PROCESSING OPTIONS contains all EMV primitive tags reserved but not used in the EMV specification, (for example ‘9F0C’, ‘9F28’and ‘9F29’, …) in the range ‘80’ to ‘9E’ and ‘9F01’ to ‘9F4F’ with values different from zero and shall be encoded in template 77

Action: NA

Procedure: Application in LT is selected and transaction is processed with LT until approval.

Pass Criteria:

- The terminal shall process the transaction until completion, by requesting a TC either at First or Second GENERATE AC.
- Values of the EMV primitive tags reserved but not used in the EMV specification, (for example ‘9F28’and ‘9F29’, …) in the range ‘80’ to ‘9E’ and ‘9F01’ to ‘9F4F’received by the LT via the DOLs are set to “00’ (Note: for CDOL2 this is applicable only if a 2nd GENERATE AC has been performed)
- TVR byte 1 - bit 1, byte 2 - bit 3, byte 4 - bit 1, byte 5 – bits 4 to 1 are set to ‘0’, received at 1st GENERATE AC.

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2CA.122.12 EMV reserved Data objects processing (9)

Test No: **2CA.122.12** - *Revision* 43j - *Number of sub cases:* 3

Objective: To ensure that the terminal ignores EMV reserved Data Objects (not used by EMV).

Reference EMV 4.3: Book 3 – Annex B1 Coding of the Tag Field of BER-TLV Data Objects

Book Bulletins - Section SB 197 - Bulletin n°197, Tokenisation Data Objects – Token Requestor ID and Last 4 Digits of PAN (May 2017)

Book Bulletins - Section SB 185 - Bulletin n°185, Biometric Terminal Specification (March 2017)

Book Bulletins - Section SB 231 - Bulletin n°231, Issuer Identification Number Extended (IINE) (December 2020)

Terminal Configuration: [PSE] supported

Terminal Configuration (New format): [PSE]

Conditions:

- LT AID value is A0000000031010
- Terminal AID Value is A0000000031010
- PDOL, CDOL1, CDOL2 and TDOL of LT contains all EMV primitive tags reserved but not used in the EMV specification, (for example ‘9F28’and ‘9F29’, …) in the range ‘80’ to ‘9E’ and ‘9F01’ to ‘9F4F’.
- All Data Elements listed in the Specification Bulletin SB 185, Table 34 are excluded from the below lists.
- All Data Elements listed in the Specification Bulletin SB 197, Table 34 are excluded from the below lists.
- All Data Elements listed in the Specification Bulletin SB 231, Table 34 are excluded from the below lists.
- PSE contains a directory file with entries for ADF.
- Case 01: FCI Issuer discretionary Data (tag BF0C) of PSE contains additionally all EMV primitive tags reserved but not used in the EMV specification, (for example ‘9F28’and ‘9F29’, …) in the range ‘80’ to ‘9E’ and ‘9F01’ to ‘9F4F’' with values different from zero.
- Case 02: FCI (tag 6F) of PSE contains additionally all EMV primitive tags reserved but not used in the EMV specification, (for example ‘9F28’and ‘9F29’, …) in the range ‘80’ to ‘9E’ and ‘9F01’ to ‘9F4F’ with values different from zero.
- Case 03: Template '73' of ADF Directory Entry contains: additionally all EMV primitive tags reserved but not used in the EMV specification, (for example ‘9F28’and ‘9F29’, …) in the range ‘80’ to ‘9E’ and ‘9F01’ to ‘9F4F’ with values different from zero.

Action: NA

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2CA.122.12 EMV reserved Data objects processing (9)

Procedure: Application in LT is selected and transaction is processed with LT until approval.

Pass Criteria:

- The terminal shall process the transaction until completion, by requesting a TC.
- Values of the EMV primitive tags reserved but not used in the EMV specification, (for example ‘9F28’and ‘9F29’, …) in the range ‘80’ to ‘9E’ and ‘9F01’ to ‘9F4F’received by the LT via the DOLs are set to “00’ (Note: for CDOL2 this is applicable only if a 2nd GENERATE AC has been performed)

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2CA.122.13 EMV reserved Data objects processing (10)

Test No: **2CA.122.13** – *Revision* 43j – *Number of sub cases:* 5

Objective: To ensure that the terminal ignores EMV reserved Data Objects (not used by EMV).

Reference EMV 4.3: Book 3 – Annex B1 Coding of the Tag Field of BER-TLV Data Objects

Book Bulletins - Section SB 185 - Bulletin n°185, Biometric Terminal Specification (March 2017)

Terminal Configuration: [Biometric] not supported

Terminal Configuration (New format): NOT[Biometric]

Conditions:

- LT AID value is A0000000031010
- Terminal AID Value is A0000000031010
- PDOL, CDOL1, CDOL2 and TDOL of LT contains EMV primitive tags reserved but not used in the EMV specification: ‘9F30’ and ‘9F31’
- Case 01: FCI (tag 6F) of ADF contains the following EMV primitive tags reserved but not used in the EMV specification: ‘9F30’ and ‘9F31’ with values different from zero within the FCI template
- Case 02: FCI Issuer discretionary Data (tag BF0C) of ADF contains the following EMV primitive tags reserved but not used in the EMV specification: ‘9F30’ and ‘9F31’ with values different from zero
- Case 03: Records contains the following EMV primitive tags reserved but not used in the EMV specification: ‘9F30’ and ‘9F31’ with values different from zero.
- Case 04: Response to GET PROCESSING OPTIONS contains the following EMV primitive tags reserved but not used in the EMV specification: ‘9F30’ and ‘9F31’ with values different from zero encoded in template 77
- Case 05: Response to GENERATE AC contains the following EMV primitive tags reserved but not used in the EMV specification: ‘9F30’ and ‘9F31’ with values different from zero encoded in template 77

Action: NA

Procedure: Application in LT is selected and transaction is processed with LT until approval.

Pass Criteria:

- The terminal shall process the transaction until completion, by requesting a TC.
- Values of the reserved tags '9F30' and '9F31' received by the LT via the DOLs are set to “00’ (Note: for CDOL2 this is applicable only if a 2nd GENERATE AC has been performed)

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2CA.122.14 EMV reserved Data objects processing (11)

Test No: **2CA.122.14** – *Revision* 43j – *Number of sub cases:* 5

Objective: To ensure that the terminal ignores EMV reserved Data Objects (not used by EMV).

Reference EMV 4.3: Book 3 – Annex B1 Coding of the Tag Field of BER-TLV Data Objects

Book Bulletins - Section SB 197 - Bulletin n°197, Tokenisation Data Objects – Token Requestor ID and Last 4 Digits of PAN (May 2017)

Book Bulletins - Section SB 185 - Bulletin n°185, Biometric Terminal Specification (March 2017)

Book Bulletins - Section SB 231 - Bulletin n°231, Issuer Identification Number Extended (IINE) (December 2020)

Terminal Configuration:

Terminal Configuration (New format): [NA]

Conditions:

- LT AID value is A0000000041010
- Terminal AID Value is A0000000041010
- PDOL, CDOL1, CDOL2 and TDOL of LT contains all EMV primitive tags reserved but not used in the EMV specification, (for example '9F28'and '9F29', …) in the range '80' to '9E' and '9F01' to '9F4F'.
- All Data Elements listed in the Specification Bulletin SB 185, Table 34 are excluded from the below lists.
- All Data Elements listed in the Specification Bulletin SB 197, Table 34 are excluded from the below lists.
- All Data Elements listed in the Specification Bulletin SB 231, Table 34 are excluded from the below lists.
- Case 01: FCI (tag 6F) of ADF contains additionally all EMV primitive tags reserved but not used in the EMV specification, (for example '9F28'and '9F29', …) in the range '80' to '9E' and '9F01' to '9F4F' with values different from zero within the FCI template
- Case 02: FCI Issuer discretionary Data (tag BF0C) of ADF contains additionally all EMV primitive tags reserved but not used in the EMV specification, (for example '9F28'and '9F29', …) in the range '80' to '9E' and '9F01' to '9F4F' with values different from zero
- Case 03: Response to GENERATE AC contains all EMV primitive tags reserved but not used in the EMV specification, (for example '9F28'and '9F29', …) in the range '80' to '9E' and '9F01' to '9F4F' with values different from zero and shall be encoded in template 77
- Case 04: Records contains: all EMV primitive tags reserved but not used in the EMV specification, (for example '9F28'and '9F29', …) in

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2CA.122.14 EMV reserved Data objects processing (11)

the range ‘80’ to ‘9E’ and ‘9F01’ to ‘9F4F’ with values different from zero (splitted over SFI 1 to 10).

- Case 05: Response to GET PROCESSING OPTIONS contains all EMV primitive tags reserved but not used in the EMV specification, (for example ‘9F28’and ‘9F29’, …) in the range ‘80’ to ‘9E’ and ‘9F01’ to ‘9F4F’ with values different from zero and shall be encoded in template 77

Action: NA

Procedure: Application in LT is selected and transaction is processed with LT until approval.

Pass Criteria:

- The terminal shall accept the card and process the transaction until completion, by requesting a TC either at First or Second GENERATE AC.
- Values of the EMV primitive tags reserved but not used in the EMV specification, (for example ‘9F28’and ‘9F29’, …) in the range ‘80’ to ‘9E’ and ‘9F01’ to ‘9F4F’received by the LT via the DOLs are set to “00’ (Note: for CDOL2 this is applicable only if a 2nd GENERATE AC has been performed)
- TVR byte 1 - bit 1, byte 2 - bit 3, byte 4 - bit 1, byte 5 – bits 4 to 1 are set to ‘0’, received at 1st GENERATE AC.

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2CA.122.15 EMV reserved Data objects processing (12)

Test No: **2CA.122.15** - *Revision* 43j - *Number of sub cases:* 3

Objective: To ensure that the terminal ignores EMV reserved Data Objects (not used by EMV).

Reference EMV 4.3: Book 3 – Annex B1 Coding of the Tag Field of BER-TLV Data Objects

Book Bulletins - Section SB 197 - Bulletin n°197, Tokenisation Data Objects – Token Requestor ID and Last 4 Digits of PAN (May 2017)

Book Bulletins - Section SB 185 - Bulletin n°185, Biometric Terminal Specification (March 2017)

Book Bulletins - Section SB 231 - Bulletin n°231, Issuer Identification Number Extended (IINE) (December 2020)

Terminal Configuration: [PSE] supported

Terminal Configuration (New format): [PSE]

Conditions:

- LT AID value is A0000000041010
- Terminal AID Value is A0000000041010
- PDOL, CDOL1, CDOL2 and TDOL of LT contains all EMV primitive tags reserved but not used in the EMV specification, (for example ‘9F28’and ‘9F29’, …) in the range ‘80’ to ‘9E’ and ‘9F01’ to ‘9F4F’.
- All Data Elements listed in the Specification Bulletin SB 185, Table 34 are excluded from the below lists.
- All Data Elements listed in the Specification Bulletin SB 197, Table 34 are excluded from the below lists.
- All Data Elements listed in the Specification Bulletin SB 231, Table 34 are excluded from the below lists.
- PSE contains a directory file with entries for ADF.
- Case 01: FCI Issuer discretionary Data (tag BF0C) of PSE contains additionally all EMV primitive tags reserved but not used in the EMV specification, (for example ‘9F28’and ‘9F29’, …) in the range ‘80’ to ‘9E’ and ‘9F01’ to ‘9F4F’' with values different from zero.
- Case 02: FCI (tag 6F) of PSE contains additionally all EMV primitive tags reserved but not used in the EMV specification, (for example ‘9F28’and ‘9F29’, …) in the range ‘80’ to ‘9E’ and ‘9F01’ to ‘9F4F’ with values different from zero.
- Case 03: Template '73' of ADF Directory Entry contains: additionally all EMV primitive tags reserved but not used in the EMV specification, (for example ‘9F28’and ‘9F29’, …) in the range ‘80’ to ‘9E’ and ‘9F01’ to ‘9F4F’ with values different from zero.

Action: NA

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2CA.122.15 EMV reserved Data objects processing (12)

Procedure: Application in LT is selected and transaction is processed with LT until approval.

Pass Criteria:

- The terminal shall accept the card and process the transaction until completion, by requesting a TC.
- Values of the EMV primitive tags reserved but not used in the EMV specification, (for example ‘9F28’and ‘9F29’, …) in the range ‘80’ to ‘9E’ and ‘9F01’ to ‘9F4F’received by the LT via the DOLs are set to “00’ (Note: for CDOL2 this is applicable only if a 2nd GENERATE AC has been performed)

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2CA.122.16 EMV reserved Data objects processing (13)

Test No: **2CA.122.16** – *Revision* 43j – *Number of sub cases:* 5

Objective: To ensure that the terminal ignores EMV reserved Data Objects (not used by EMV).

Reference EMV 4.3: Book 3 – Annex B1 Coding of the Tag Field of BER-TLV Data Objects

Book Bulletins - Section SB 185 - Bulletin n°185, Biometric Terminal Specification (March 2017)

Terminal Configuration: [Biometric] not supported

Terminal Configuration (New format): NOT[Biometric]

Conditions:

- LT AID value is A0000000041010
- Terminal AID Value is A0000000041010
- PDOL, CDOL1, CDOL2 and TDOL of LT contains EMV primitive tags reserved but not used in the EMV specification: ‘9F30’ and ‘9F31’
- Case 01: FCI (tag 6F) of ADF contains the following EMV primitive tags reserved but not used in the EMV specification: ‘9F30’ and ‘9F31’ with values different from zero within the FCI template
- Case 02: FCI Issuer discretionary Data (tag BF0C) of ADF contains the following EMV primitive tags reserved but not used in the EMV specification: ‘9F30’ and ‘9F31’ with values different from zero
- Case 03: Records contains the following EMV primitive tags reserved but not used in the EMV specification: ‘9F30’ and ‘9F31’ with values different from zero.
- Case 04: Response to GET PROCESSING OPTIONS contains the following EMV primitive tags reserved but not used in the EMV specification: ‘9F30’ and ‘9F31’ with values different from zero encoded in template 77
- Case 05: Response to GENERATE AC contains the following EMV primitive tags reserved but not used in the EMV specification: ‘9F30’ and ‘9F31’ with values different from zero encoded in template 77

Action: NA

Procedure: Application in LT is selected and transaction is processed with LT until approval.

Pass Criteria:

- The terminal shall process the transaction until completion, by requesting a TC.
- Values of the reserved tags '9F30' and '9F31' received by the LT via the DOLs are set to “00’ (Note: for CDOL2 this is applicable only if a 2nd GENERATE AC has been performed)

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2CA.122.17 EMV reserved Data objects processing (14)

Test No: **2CA.122.17** – *Revision* 43j – *Number of sub cases:* 5

Objective: To ensure that the terminal ignores EMV reserved Data Objects (not used by EMV).

Reference EMV 4.3: Book 3 – Annex B1 Coding of the Tag Field of BER-TLV Data Objects

Book Bulletins - Section SB 197 - Bulletin n°197, Tokenisation Data Objects – Token Requestor ID and Last 4 Digits of PAN (May 2017)

Book Bulletins - Section SB 185 - Bulletin n°185, Biometric Terminal Specification (March 2017)

Book Bulletins - Section SB 231 - Bulletin n°231, Issuer Identification Number Extended (IINE) (December 2020)

Terminal Configuration:

Terminal Configuration (New format): [NA]

Conditions:

- LT AID value is A0000000651010
- Terminal AID Value is A0000000651010
- PDOL, CDOL1, CDOL2 and TDOL of LT contains all EMV primitive tags reserved but not used in the EMV specification, (for example '9F28'and '9F29', …) in the range '80' to '9E' and '9F01' to '9F4F'.
- All Data Elements listed in the Specification Bulletin SB 185, Table 34 are excluded from the below lists.
- All Data Elements listed in the Specification Bulletin SB 197, Table 34 are excluded from the below lists.
- All Data Elements listed in the Specification Bulletin SB 231, Table 34 are excluded from the below lists.
- Case 01: FCI (tag 6F) of ADF contains additionally all EMV primitive tags reserved but not used in the EMV specification, (for example '9F28'and '9F29', …) in the range '80' to '9E' and '9F01' to '9F4F' with values different from zero within the FCI template
- Case 02: FCI Issuer discretionary Data (tag BF0C) of ADF contains additionally all EMV primitive tags reserved but not used in the EMV specification, (for example '9F28'and '9F29', …) in the range '80' to '9E' and '9F01' to '9F4F' with values different from zero
- Case 03: Response to GENERATE AC contains all EMV primitive tags reserved but not used in the EMV specification, (for example '9F28'and '9F29', …) in the range '80' to '9E' and '9F01' to '9F4F' with values different from zero and shall be encoded in template 77
- Case 04: Records contains: all EMV primitive tags reserved but not used in the EMV specification, (for example '9F28'and '9F29', …) in

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the range ‘80’ to ‘9E’ and ‘9F01’ to ‘9F4F’ with values different from zero (splitted over SFI 1 to 10).

- Case 05: Response to GET PROCESSING OPTIONS contains all EMV primitive tags reserved but not used in the EMV specification, (for example ‘9F28’and ‘9F29’, …) in the range ‘80’ to ‘9E’ and ‘9F01’ to ‘9F4F’ with values different from zero and shall be encoded in template 77

Action: NA

Procedure: Application in LT is selected and transaction is processed with LT until approval.

Pass Criteria:

- The terminal shall accept the card and process the transaction until completion, by requesting a TC either at First or Second GENERATE AC.
- Values of the EMV primitive tags reserved but not used in the EMV specification, (for example ‘9F28’and ‘9F29’, …) in the range ‘80’ to ‘9E’ and ‘9F01’ to ‘9F4F’received by the LT via the DOLs are set to “00’ (Note: for CDOL2 this is applicable only if a 2nd GENERATE AC has been performed)
- TVR byte 1 - bit 1, byte 2 - bit 3, byte 4 - bit 1, byte 5 – bits 4 to 1 are set to ‘0’, received at 1st GENERATE AC.

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2CA.122.18 EMV reserved Data objects processing (15)

Test No: **2CA.122.18** - *Revision* 43j - *Number of sub cases:* 3

Objective: To ensure that the terminal ignores EMV reserved Data Objects (not used by EMV).

Reference EMV 4.3: Book 3 – Annex B1 Coding of the Tag Field of BER-TLV Data Objects

Book Bulletins - Section SB 197 - Bulletin n°197, Tokenisation Data Objects – Token Requestor ID and Last 4 Digits of PAN (May 2017)

Book Bulletins - Section SB 185 - Bulletin n°185, Biometric Terminal Specification (March 2017)

Book Bulletins - Section SB 231 - Bulletin n°231, Issuer Identification Number Extended (IINE) (December 2020)

Terminal Configuration: [PSE] supported

Terminal Configuration (New format): [PSE]

Conditions:

- LT AID value is A0000000651010
- Terminal AID Value is A0000000651010
- PDOL, CDOL1, CDOL2 and TDOL of LT contains all EMV primitive tags reserved but not used in the EMV specification, (for example ‘9F28’and ‘9F29’, …) in the range ‘80’ to ‘9E’ and ‘9F01’ to ‘9F4F’.
- All Data Elements listed in the Specification Bulletin SB 185, Table 34 are excluded from the below lists.
- All Data Elements listed in the Specification Bulletin SB 197, Table 34 are excluded from the below lists.
- All Data Elements listed in the Specification Bulletin SB 231, Table 34 are excluded from the below lists.
- PSE contains a directory file with entries for ADF.
- Case 01: FCI Issuer discretionary Data (tag BF0C) of PSE contains additionally all EMV primitive tags reserved but not used in the EMV specification, (for example ‘9F28’and ‘9F29’, …) in the range ‘80’ to ‘9E’ and ‘9F01’ to ‘9F4F’' with values different from zero.
- Case 02: FCI (tag 6F) of PSE contains additionally all EMV primitive tags reserved but not used in the EMV specification, (for example ‘9F28’and ‘9F29’, …) in the range ‘80’ to ‘9E’ and ‘9F01’ to ‘9F4F’ with values different from zero.
- Case 03: Template '73' of ADF Directory Entry contains: additionally all EMV primitive tags reserved but not used in the EMV specification, (for example ‘9F28’and ‘9F29’, …) in the range ‘80’ to ‘9E’ and ‘9F01’ to ‘9F4F’ with values different from zero.

Action: NA

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2CA.122.18 EMV reserved Data objects processing (15)

Procedure: Application in LT is selected and transaction is processed with LT until approval.

Pass Criteria:

- The terminal shall accept the card and process the transaction until completion, by requesting a TC.
- Values of the EMV primitive tags reserved but not used in the EMV specification, (for example ‘9F28’and ‘9F29’, …) in the range ‘80’ to ‘9E’ and ‘9F01’ to ‘9F4F’received by the LT via the DOLs are set to “00’ (Note: for CDOL2 this is applicable only if a 2nd GENERATE AC has been performed)

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2CA.122.19 EMV reserved Data objects processing (16)

Test No: **2CA.122.19** – *Revision* 43j – *Number of sub cases:* 5

Objective: To ensure that the terminal ignores EMV reserved Data Objects (not used by EMV).

Reference EMV 4.3: Book 3 – Annex B1 Coding of the Tag Field of BER-TLV Data Objects

Book Bulletins - Section SB 185 - Bulletin n°185, Biometric Terminal Specification (March 2017)

Terminal Configuration: [Biometric] not supported

Terminal Configuration (New format): NOT[Biometric]

Conditions:

- LT AID value is A0000000651010
- Terminal AID Value is A0000000651010
- PDOL, CDOL1, CDOL2 and TDOL of LT contains EMV primitive tags reserved but not used in the EMV specification: ‘9F30’ and ‘9F31’
- Case 01: FCI (tag 6F) of ADF contains the following EMV primitive tags reserved but not used in the EMV specification: ‘9F30’ and ‘9F31’ with values different from zero within the FCI template
- Case 02: FCI Issuer discretionary Data (tag BF0C) of ADF contains the following EMV primitive tags reserved but not used in the EMV specification: ‘9F30’ and ‘9F31’ with values different from zero
- Case 03: Records contains the following EMV primitive tags reserved but not used in the EMV specification: ‘9F30’ and ‘9F31’ with values different from zero.
- Case 04: Response to GET PROCESSING OPTIONS contains the following EMV primitive tags reserved but not used in the EMV specification: ‘9F30’ and ‘9F31’ with values different from zero encoded in template 77
- Case 05: Response to GENERATE AC contains the following EMV primitive tags reserved but not used in the EMV specification: ‘9F30’ and ‘9F31’ with values different from zero encoded in template 77

Action: NA

Procedure: Application in LT is selected and transaction is processed with LT until approval.

Pass Criteria:

- The terminal shall process the transaction until completion, by requesting a TC.
- Values of the reserved tags '9F30' and '9F31' received by the LT via the DOLs are set to “00’ (Note: for CDOL2 this is applicable only if a 2nd GENERATE AC has been performed)

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2CA.122.20 EMV reserved Data objects processing (17)

Test No: **2CA.122.20** – *Revision* 43j – *Number of sub cases:* 5

Objective: To ensure that the terminal ignores EMV reserved Data Objects (not used by EMV).

Reference EMV 4.3: Book 3 – Annex B1 Coding of the Tag Field of BER-TLV Data Objects

Book Bulletins - Section SB 197 - Bulletin n°197, Tokenisation Data Objects – Token Requestor ID and Last 4 Digits of PAN (May 2017)

Book Bulletins - Section SB 185 - Bulletin n°185, Biometric Terminal Specification (March 2017)

Book Bulletins - Section SB 231 - Bulletin n°231, Issuer Identification Number Extended (IINE) (December 2020)

Terminal Configuration:

Terminal Configuration (New format): [NA]

Conditions:

- LT AID value is A000000333010101
- Terminal AID Value is A000000333010101
- PDOL, CDOL1, CDOL2 and TDOL of LT contains all EMV primitive tags reserved but not used in the EMV specification, (for example ‘9F28’and ‘9F29’, …) in the range ‘80’ to ‘9E’ and ‘9F01’ to ‘9F4F’.
- All Data Elements listed in the Specification Bulletin SB 185, Table 34 are excluded from the below lists.
- All Data Elements listed in the Specification Bulletin SB 197, Table 34 are excluded from the below lists.
- All Data Elements listed in the Specification Bulletin SB 231, Table 34 are excluded from the below lists.
- Case 01: FCI (tag 6F) of ADF contains additionally all EMV primitive tags reserved but not used in the EMV specification, (for example ‘9F28’and ‘9F29’, …) in the range ‘80’ to ‘9E’ and ‘9F01’ to ‘9F4F’ with values different from zero within the FCI template
- Case 02: FCI Issuer discretionary Data (tag BF0C) of ADF contains additionally all EMV primitive tags reserved but not used in the EMV specification, (for example ‘9F28’and ‘9F29’, …) in the range ‘80’ to ‘9E’ and ‘9F01’ to ‘9F4F’ with values different from zero
- Case 03: Response to GENERATE AC contains all EMV primitive tags reserved but not used in the EMV specification, (for example ‘9F28’and ‘9F29’, …) in the range ‘80’ to ‘9E’ and ‘9F01’ to ‘9F4F’ with values different from zero and shall be encoded in template 77
- Case 04: Records contains: all EMV primitive tags reserved but not used in the EMV specification, (for example ‘9F28’and ‘9F29’, …) in

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2CA.122.20 EMV reserved Data objects processing (17)

the range ‘80’ to ‘9E’ and ‘9F01’ to ‘9F4F’ with values different from zero (splitted over SFI 1 to 10).

- Case 05: Response to GET PROCESSING OPTIONS contains all EMV primitive tags reserved but not used in the EMV specification, (for example ‘9F28’and ‘9F29’, …) in the range ‘80’ to ‘9E’ and ‘9F01’ to ‘9F4F’ with values different from zero and shall be encoded in template 77

Action: NA

Procedure: Application in LT is selected and transaction is processed with LT until approval.

Pass Criteria:

- The terminal shall accept the card and process the transaction until completion, by requesting a TC either at First or Second GENERATE AC.
- Values of the EMV primitive tags reserved but not used in the EMV specification, (for example ‘9F28’and ‘9F29’, …) in the range ‘80’ to ‘9E’ and ‘9F01’ to ‘9F4F’received by the LT via the DOLs are set to “00’ (Note: for CDOL2 this is applicable only if a 2nd GENERATE AC has been performed)
- TVR byte 1 - bit 1, byte 2 - bit 3, byte 4 - bit 1, byte 5 – bits 4 to 1 are set to ‘0’, received at 1st GENERATE AC.

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2CA.122.21 EMV reserved Data objects processing (18)

Test No: **2CA.122.21** - *Revision* 43j - *Number of sub cases:* 3

Objective: To ensure that the terminal ignores EMV reserved Data Objects (not used by EMV).

Reference EMV 4.3: Book 3 – Annex B1 Coding of the Tag Field of BER-TLV Data Objects

Book Bulletins - Section SB 197 - Bulletin n°197, Tokenisation Data Objects – Token Requestor ID and Last 4 Digits of PAN (May 2017)

Book Bulletins - Section SB 185 - Bulletin n°185, Biometric Terminal Specification (March 2017)

Book Bulletins - Section SB 231 - Bulletin n°231, Issuer Identification Number Extended (IINE) (December 2020)

Terminal Configuration: [PSE] supported

Terminal Configuration (New format): [PSE]

Conditions:

- LT AID value is A000000333010101
- Terminal AID Value is A000000333010101
- PDOL, CDOL1, CDOL2 and TDOL of LT contains all EMV primitive tags reserved but not used in the EMV specification, (for example ‘9F28’and ‘9F29’, …) in the range ‘80’ to ‘9E’ and ‘9F01’ to ‘9F4F’.
- All Data Elements listed in the Specification Bulletin SB 185, Table 34 are excluded from the below lists.
- All Data Elements listed in the Specification Bulletin SB 197, Table 34 are excluded from the below lists.
- All Data Elements listed in the Specification Bulletin SB 231, Table 34 are excluded from the below lists.
- PSE contains a directory file with entries for ADF.
- Case 01: FCI Issuer discretionary Data (tag BF0C) of PSE contains additionally all EMV primitive tags reserved but not used in the EMV specification, (for example ‘9F28’and ‘9F29’, …) in the range ‘80’ to ‘9E’ and ‘9F01’ to ‘9F4F’' with values different from zero.
- Case 02: FCI (tag 6F) of PSE contains additionally all EMV primitive tags reserved but not used in the EMV specification, (for example ‘9F28’and ‘9F29’, …) in the range ‘80’ to ‘9E’ and ‘9F01’ to ‘9F4F’ with values different from zero.
- Case 03: Template '73' of ADF Directory Entry contains: additionally all EMV primitive tags reserved but not used in the EMV specification, (for example ‘9F28’and ‘9F29’, …) in the range ‘80’ to ‘9E’ and ‘9F01’ to ‘9F4F’ with values different from zero.

Action: NA

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2CA.122.21 EMV reserved Data objects processing (18)

Procedure: Application in LT is selected and transaction is processed with LT until approval.

Pass Criteria:

- The terminal shall accept the card and process the transaction until completion, by requesting a TC.
- Values of the EMV primitive tags reserved but not used in the EMV specification, (for example ‘9F28’and ‘9F29’, …) in the range ‘80’ to ‘9E’ and ‘9F01’ to ‘9F4F’received by the LT via the DOLs are set to “00’ (Note: for CDOL2 this is applicable only if a 2nd GENERATE AC has been performed)

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2CA.122.22 EMV reserved Data objects processing (19)

Test No: **2CA.122.22** – *Revision* 43j – *Number of sub cases:* 5

Objective: To ensure that the terminal ignores EMV reserved Data Objects (not used by EMV).

Reference EMV 4.3: Book 3 – Annex B1 Coding of the Tag Field of BER-TLV Data Objects

Book Bulletins - Section SB 185 - Bulletin n°185, Biometric Terminal Specification (March 2017)

Terminal Configuration: [Biometric] not supported

Terminal Configuration (New format): NOT[Biometric]

Conditions:

- LT AID value is A000000333010101
- Terminal AID Value is A000000333010101
- PDOL, CDOL1, CDOL2 and TDOL of LT contains EMV primitive tags reserved but not used in the EMV specification: ‘9F30’ and ‘9F31’
- Case 01: FCI (tag 6F) of ADF contains the following EMV primitive tags reserved but not used in the EMV specification: ‘9F30’ and ‘9F31’ with values different from zero within the FCI template
- Case 02: FCI Issuer discretionary Data (tag BF0C) of ADF contains the following EMV primitive tags reserved but not used in the EMV specification: ‘9F30’ and ‘9F31’ with values different from zero
- Case 03: Records contains the following EMV primitive tags reserved but not used in the EMV specification: ‘9F30’ and ‘9F31’ with values different from zero.
- Case 04: Response to GET PROCESSING OPTIONS contains the following EMV primitive tags reserved but not used in the EMV specification: ‘9F30’ and ‘9F31’ with values different from zero encoded in template 77
- Case 05: Response to GENERATE AC contains the following EMV primitive tags reserved but not used in the EMV specification: ‘9F30’ and ‘9F31’ with values different from zero encoded in template 77

Action: NA

Procedure: Application in LT is selected and transaction is processed with LT until approval.

Pass Criteria:

- The terminal shall process the transaction until completion, by requesting a TC.
- Values of the reserved tags '9F30' and '9F31' received by the LT via the DOLs are set to “00’ (Note: for CDOL2 this is applicable only if a 2nd GENERATE AC has been performed)

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2CA.122.23 EMV reserved Data objects processing (20)

Test No: **2CA.122.23** – *Revision* 43j – *Number of sub cases:* 5

Objective: To ensure that the terminal ignores EMV reserved Data Objects (not used by EMV).

Reference EMV 4.3: Book 3 – Annex B1 Coding of the Tag Field of BER-TLV Data Objects

Book Bulletins - Section SB 197 - Bulletin n°197, Tokenisation Data Objects – Token Requestor ID and Last 4 Digits of PAN (May 2017)

Book Bulletins - Section SB 185 - Bulletin n°185, Biometric Terminal Specification (March 2017)

Book Bulletins - Section SB 231 - Bulletin n°231, Issuer Identification Number Extended (IINE) (December 2020)

Terminal Configuration:

Terminal Configuration (New format): [NA]

Conditions:

- LT AID value is A0000001523010
- Terminal AID Value is A0000001523010
- PDOL, CDOL1, CDOL2 and TDOL of LT contains all EMV primitive tags reserved but not used in the EMV specification, (for example '9F28'and '9F29', …) in the range '80' to '9E' and '9F01' to '9F4F'.
- All Data Elements listed in the Specification Bulletin SB 185, Table 34 are excluded from the below lists.
- All Data Elements listed in the Specification Bulletin SB 197, Table 34 are excluded from the below lists.
- All Data Elements listed in the Specification Bulletin SB 231, Table 34 are excluded from the below lists.
- Case 01: FCI (tag 6F) of ADF contains additionally all EMV primitive tags reserved but not used in the EMV specification, (for example '9F28'and '9F29', …) in the range '80' to '9E' and '9F01' to '9F4F' with values different from zero within the FCI template
- Case 02: FCI Issuer discretionary Data (tag BF0C) of ADF contains additionally all EMV primitive tags reserved but not used in the EMV specification, (for example '9F28'and '9F29', …) in the range '80' to '9E' and '9F01' to '9F4F' with values different from zero
- Case 03: Response to GENERATE AC contains all EMV primitive tags reserved but not used in the EMV specification, (for example '9F28'and '9F29', …) in the range '80' to '9E' and '9F01' to '9F4F' with values different from zero and shall be encoded in template 77
- Case 04: Records contains: all EMV primitive tags reserved but not used in the EMV specification, (for example '9F28'and '9F29', …) in

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the range ‘80’ to ‘9E’ and ‘9F01’ to ‘9F4F’ with values different from zero (splitted over SFI 1 to 10).

- Case 05: Response to GET PROCESSING OPTIONS contains all EMV primitive tags reserved but not used in the EMV specification, (for example ‘9F28’and ‘9F29’, …) in the range ‘80’ to ‘9E’ and ‘9F01’ to ‘9F4F’ with values different from zero and shall be encoded in template 77

Action: NA

Procedure: Application in LT is selected and transaction is processed with LT until approval.

Pass Criteria:

- The terminal shall accept the card and process the transaction until completion, by requesting a TC either at First or Second GENERATE AC.
- Values of the EMV primitive tags reserved but not used in the EMV specification, (for example ‘9F28’and ‘9F29’, …) in the range ‘80’ to ‘9E’ and ‘9F01’ to ‘9F4F’received by the LT via the DOLs are set to “00’ (Note: for CDOL2 this is applicable only if a 2nd GENERATE AC has been performed)
- TVR byte 1 - bit 1, byte 2 - bit 3, byte 4 - bit 1, byte 5 – bits 4 to 1 are set to ‘0’, received at 1st GENERATE AC.

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2CA.122.24 EMV reserved Data objects processing (21)

Test No: **2CA.122.24** - *Revision* 43j - *Number of sub cases:* 3

Objective: To ensure that the terminal ignores EMV reserved Data Objects (not used by EMV).

Reference EMV 4.3: Book 3 – Annex B1 Coding of the Tag Field of BER-TLV Data Objects

Book Bulletins - Section SB 197 - Bulletin n°197, Tokenisation Data Objects – Token Requestor ID and Last 4 Digits of PAN (May 2017)

Book Bulletins - Section SB 185 - Bulletin n°185, Biometric Terminal Specification (March 2017)

Book Bulletins - Section SB 231 - Bulletin n°231, Issuer Identification Number Extended (IINE) (December 2020)

Terminal Configuration: [PSE] supported

Terminal Configuration (New format): [PSE]

Conditions:

- LT AID value is A0000001523010
- Terminal AID Value is A0000001523010
- PDOL, CDOL1, CDOL2 and TDOL of LT contains all EMV primitive tags reserved but not used in the EMV specification, (for example ‘9F28’and ‘9F29’, …) in the range ‘80’ to ‘9E’ and ‘9F01’ to ‘9F4F’.
- All Data Elements listed in the Specification Bulletin SB 185, Table 34 are excluded from the below lists.
- All Data Elements listed in the Specification Bulletin SB 197, Table 34 are excluded from the below lists.
- All Data Elements listed in the Specification Bulletin SB 231, Table 34 are excluded from the below lists.
- PSE contains a directory file with entries for ADF.
- Case 01: FCI Issuer discretionary Data (tag BF0C) of PSE contains additionally all EMV primitive tags reserved but not used in the EMV specification, (for example ‘9F28’and ‘9F29’, …) in the range ‘80’ to ‘9E’ and ‘9F01’ to ‘9F4F’' with values different from zero.
- Case 02: FCI (tag 6F) of PSE contains additionally all EMV primitive tags reserved but not used in the EMV specification, (for example ‘9F28’and ‘9F29’, …) in the range ‘80’ to ‘9E’ and ‘9F01’ to ‘9F4F’ with values different from zero.
- Case 03: Template '73' of ADF Directory Entry contains: additionally all EMV primitive tags reserved but not used in the EMV specification, (for example ‘9F28’and ‘9F29’, …) in the range ‘80’ to ‘9E’ and ‘9F01’ to ‘9F4F’ with values different from zero.

Action: NA

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2CA.122.24 EMV reserved Data objects processing (21)

Procedure: Application in LT is selected and transaction is processed with LT until approval.

Pass Criteria:

- The terminal shall accept the card and process the transaction until completion, by requesting a TC.
- Values of the EMV primitive tags reserved but not used in the EMV specification, (for example ‘9F28’and ‘9F29’, …) in the range ‘80’ to ‘9E’ and ‘9F01’ to ‘9F4F’received by the LT via the DOLs are set to “00’ (Note: for CDOL2 this is applicable only if a 2nd GENERATE AC has been performed)

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2CA.122.25 EMV reserved Data objects processing (22)

Test No: **2CA.122.25** – *Revision* 43j – *Number of sub cases:* 5

Objective: To ensure that the terminal ignores EMV reserved Data Objects (not used by EMV).

Reference EMV 4.3: Book 3 – Annex B1 Coding of the Tag Field of BER-TLV Data Objects

Book Bulletins - Section SB 185 - Bulletin n°185, Biometric Terminal Specification (March 2017)

Terminal Configuration: [Biometric] not supported

Terminal Configuration (New format): NOT[Biometric]

Conditions:

- LT AID value is A0000001523010
- Terminal AID Value is A0000001523010
- PDOL, CDOL1, CDOL2 and TDOL of LT contains EMV primitive tags reserved but not used in the EMV specification: ‘9F30’ and ‘9F31’
- Case 01: FCI (tag 6F) of ADF contains the following EMV primitive tags reserved but not used in the EMV specification: ‘9F30’ and ‘9F31’ with values different from zero within the FCI template
- Case 02: FCI Issuer discretionary Data (tag BF0C) of ADF contains the following EMV primitive tags reserved but not used in the EMV specification: ‘9F30’ and ‘9F31’ with values different from zero
- Case 03: Records contains the following EMV primitive tags reserved but not used in the EMV specification: ‘9F30’ and ‘9F31’ with values different from zero.
- Case 04: Response to GET PROCESSING OPTIONS contains the following EMV primitive tags reserved but not used in the EMV specification: ‘9F30’ and ‘9F31’ with values different from zero encoded in template 77
- Case 05: Response to GENERATE AC contains the following EMV primitive tags reserved but not used in the EMV specification: ‘9F30’ and ‘9F31’ with values different from zero encoded in template 77

Action: NA

Procedure: Application in LT is selected and transaction is processed with LT until approval.

Pass Criteria:

- The terminal shall process the transaction until completion, by requesting a TC.
- Values of the reserved tags '9F30' and '9F31' received by the LT via the DOLs are set to “00’ (Note: for CDOL2 this is applicable only if a 2nd GENERATE AC has been performed)

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2CA.124.00 Application Version Number assigned by the payment system

Test No: **2CA.124.00** - *Revision* 43j - *Number of sub cases:* 6

Objective: To ensure that the terminal maintains the Application Dependent Data for each supported AID(s) assigned by the payment system.

Reference EMV 4.3: Book 4 - Section 10.2 - Application Dependent Data

Terminal Configuration: NA

Terminal Configuration (New format): [NA]

Conditions:

- Terminal is configured at the same time with 'NbAIDsForTest' AIDs and each of them have a set of different data according to table 7 (excluding the following data elements managed at the Terminal Level: Transaction Currency Code, Transaction Currency Exponent, Transaction Reference Currency Code and Transaction Reference Currency Exponent) of section 10.2, Book 4 and for each AID the data values are different.
    - The number of AIDs 'NbAIDsForTest' configured in the Terminal depends on the response of the ICS question: 'How Many AIDs can be supported'= NbMaxAIDs, where NbAIDsForTest=MIN(11, NbMaxAIDs).
- For AID 1: Terminal Action Code – Default, Denial and Online are set to 0
- For AID 2 Terminal Action Code – Default, and Online are set to '0' Terminal Action Code – Denial are set to 00 80 00 00 00.
- For AID 3 Terminal Action Code – Default, and Online are set to '0' Terminal Action Code – Denial are set to 00 00 40 00 00.
- For AID 4 Terminal Action Code – Default, and Online are set to '0' Terminal Action Code – Denial are set to 0.
- For AID 5 Terminal Action Code – Default, and Online are set to '0' Terminal Action Code – Denial are set to 00 20 00 00 00.
- For AID 6 Terminal Action Code – Default, and Online are set to '0' Terminal Action Code – Denial are set to 20 00 00 00 00.
- For AID 7 Terminal Action Code – Default, and Online are set to '0' Terminal Action Code – Denial are set to 00 40 00 00 00.
- For AID 8 Terminal Action Code – Default, and Online are set to '0' Terminal Action Code – Denial are set to 0.
- For AID 9 Terminal Action Code – Default, and Online are set to '0' Terminal Action Code – Denial are set to 00 80 00 00 00.
- For AID 10 Terminal Action Code – Default, and Online are set to '0' Terminal Action Code – Denial are set to 00 60 00 00 00.

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2CA.124.00 Application Version Number assigned by the payment system

- For AID 11: Terminal Action Code – Default, Denial and Online are set to 0
- CDOL1 requests all data element identified with Tags of the Table 7 except AID and CA PK Index (9F22): Acquirer Identifier, Merchant Category Code, Merchant Identifier, Merchant Name and Location, Terminal Floor Limit, Terminal Identification and also the TVR.
- The below subcases are run depending on the number of AID supported, if less than 11 AIDs are supported, the corresponding number of subcases are run
- Case 01: AID 1 is selected
- Case 02: AID 2 is selected and ICC Application Version Number does not match Terminal Application Version Number.
- Case 03: AID 3 is selected and CVM list contain only one rule which is an 'Unrecognised CVM'.
- Case 04: AID 4 is selected.
- Case 05: AID 5 is selected and Application is not yet effective in Card.
- Case 06: AID 6 is selected and CVM list not present in Card and AIP indicates cardholder verification is supported.
- Case 07: AID 7 is selected, and Application is expired in Card
- Case 08: AID 08 is selected.
- Case 09: AID 9 is selected and ICC Application Version Number matches Terminal Application Version Number.
- Case 10: AID 10 is selected and Application is effective and not expired in Card.
- Case 11: AID 11 is selected

Action: NA

Procedure: Application in LT is selected and transaction is processed with LT.

Pass Criteria:

- LT shall received the appropriate unique value of each data from CDOL1 for the selected AID.
- Case 01 04, 08, 09, 10 and 11: The terminal shall process the transaction until completion with Approval.
- Case 02, 03, 05, 06 and 07: The terminal shall process the transaction until completion with Decline.

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2CA.124.01 Application Version Number assigned by the payment system

Test No: **2CA.124.01** - *Revision* 43j - *Number of sub cases:* 6

Objective: To ensure that the terminal maintains the Application Dependent Data for each supported AID(s) assigned by the payment system.

Reference EMV 4.3: Book 4 - Section 10.2 - Application Dependent Data

Terminal Configuration: [Offline Only] supported

Terminal Configuration (New format): [Offline Only]

Conditions:

- Terminal is configured <u>at the same time</u> with 'NbAIDsForTest' different AIDs and each of them have a set of different data according to table 7 (excluding the following data elements managed at the Terminal Level: Transaction Currency Code, Transaction Currency Exponent, Transaction Reference Currency Code and Transaction Reference Currency Exponent) of section 10.2, Book 4 and for each AID the data values are different.
    - The number of AIDs 'NbAIDsForTest' configured in the Terminal depends on the response of the ICS question: 'How Many AIDs can be supported'= NbMaxAIDs, where NbAIDsForTest=MIN(11, NbMaxAIDs).
- For AID 1: Terminal Action Code – Default, Denial and Online are set to 0
- For AID 2: Terminal Action Code – Online, and Denial are set to '0' Terminal Action Code – Default are set to 20 00 00 00 00.
- For AID 3: Terminal Action Code – Online, and Denial are set to '0' Terminal Action Code – Default are set to 00 80 00 00 00.
- For AID 4: Terminal Action Code – Online, and Denial are set to '0' Terminal Action Code – Default are set to 00 00 40 00 00.
- For AID 5: Terminal Action Code – Online, and Denial are set to '0' Terminal Action Code – Default are set to 00 40 00 00 00.
- For AID 6: Terminal Action Code – Online, and Denial are set to '0' Terminal Action Code – Default are set to 00 20 00 00 00.
- For AID 7 Terminal Action Code – Online, and Denial are set to '0' Terminal Action Code – Default are set to 00 40 00 00 00.
- For AID 8 Terminal Action Code – Online, and Denial are set to '0' Terminal Action Code – Default are set to 0.
- For AID 9 Terminal Action Code – Online, and Denial are set to '0' Terminal Action Code – Default are set to 00 80 00 00 00.
- For AID 10 Terminal Action Code –Online, and Denial are set to '0' Terminal Action Code – Default are set to 00 60 00 00 00.

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2CA.124.01 Application Version Number assigned by the payment system

- For AID 11: Terminal Action Code – Default, Denial and Online are set to 0
- The below subcases are run depending on the number of AID supported, if less than 11 AIDs are supported, the corresponding number of subcases are run
- Case 01: AID 1 is selected
- Case 02: AID 2 is selected and CVM List not present in Card and AIP indicates cardholder verification is supported.
- Case 03: AID 3 is selected and ICC Application Version Number does not match Terminal Application Version Number.
- Case 04: AID 4 is selected and CVM list contain only one rules which is an 'Unrecognised CVM'.
- Case 05: AID 5 is selected and Application is expired in Card.
- Case 06: AID 6 is selected and Application is not yet effective in Card.
- Case 07: AID 7 is selected, and Application is expired in Card
- Case 08: AID 08 is selected.
- Case 09: AID 9 is selected and ICC Application Version Number matches Terminal Application Version Number.
- Case 10: AID 10 is selected and Application is effective and not expired in Card.
- Case 11: AID 11 is selected

Action: NA

Procedure: Application in LT is selected and transaction is processed with LT.

Pass Criteria:

- Case 01, 08, 09, 10 and 11: The terminal shall process the transaction until completion with Approval.
- Case 02 to 07: The terminal shall process the transaction until completion with Decline

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2CA.124.02 Application Version Number assigned by the payment system

Test No: **2CA.124.02** - *Revision* 43j - *Number of sub cases:* 6

Objective: To ensure that the terminal maintains the Application Dependent Data for each supported AID(s) assigned by the payment system.

Reference EMV 4.3: Book 4 - Section 10.2 - Application Dependent Data

Terminal Configuration: [Offline/Online Capable] supported

Terminal Configuration (New format): [Offline/Online Capable]

Conditions:

- Terminal is configured <u>at the same time</u> with 'NbAIDsForTest' different AIDs and each of them have a set of different data according to table 7 (excluding the following data elements managed at the Terminal Level: Transaction Currency Code, Transaction Currency Exponent, Transaction Reference Currency Code and Transaction Reference Currency Exponent) of section 10.2, Book 4 and for each AID the data values are different.
    - The number of AIDs 'NbAIDsForTest' configured in the Terminal depends on the response of the ICS question: 'How Many AIDs can be supported'= NbMaxAIDs, where NbAIDsForTest=MIN(11, NbMaxAIDs).
- For AID 1: Terminal Action Code – Default, Denial and Online are set to 0
- For AID 2: Terminal Action Code – Default, and Denial are set to '0' Terminal Action Code – Online are set to 00 20 00 00 00.
- For AID 3: Terminal Action Code – Default, and Denial are set to '0' Terminal Action Code – Online are set to 20 00 00 00 00.
- For AID 4: Terminal Action Code – Default, and Denial are set to '0' Terminal Action Code – Online are set to 00 80 00 00 00.
- For AID 5: Terminal Action Code – Default, and Denial are set to '0' Terminal Action Code – Online are set to 00 00 40 00 00.
- For AID 6: Terminal Action Code – Default, and Denial are set to '0' Terminal Action Code – Online are set to 00 40 00 00 00.
- For AID 7 Terminal Action Code – Default, and Denial are set to '0' Terminal Action Code – Online are set to 00 40 00 00 00.
- For AID 8 Terminal Action Code – Default, and Denial are set to '0' Terminal Action Code – Online are set to 0.
- For AID 9 Terminal Action Code – Default, and Denial are set to '0' Terminal Action Code – Online are set to 00 80 00 00 00.
- For AID 10 Terminal Action Code – Default, and Denial are set to '0' Terminal Action Code – Online are set to 00 60 00 00 00.

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2CA.124.02 Application Version Number assigned by the payment system

- For AID 11: Terminal Action Code – Default, Denial and Online are set to 0
- The below subcases are run depending on the number of AID supported, if less than 11 AIDs are supported, the corresponding number of subcases are run
- Case 01: AID 1 is selected
- Case 02: AID 2 is selected and Application is not yet effective in Card.
- Case 03: AID 3 is selected and CVM List not present in Card and AIP indicates cardholder verification is supported.
- Case 04: AID 4 is selected and ICC Application Version Number does not match Terminal Application Version Number.
- Case 05: AID 5 is selected and CVM list contain only one rules which is an 'Unrecognised CVM'.
- Case 06: AID 6 is selected and Application is expired in Card.
- Case 07: AID 7 is selected, and Application is expired in Card
- Case 08: AID 08 is selected.
- Case 09: AID 9 is selected and ICC Application Version Number matches Terminal Application Version Number.
- Case 10: AID 10 is selected and Application is effective and not expired in Card.
- Case 11: AID 11 is selected

Action: NA

Procedure: Application in LT is selected and transaction is processed with LT.

Pass Criteria:

- Case 01, 08, 09, 10 and 11: The terminal shall process the transaction offline until completion with Approval.
- Case 02 to 07: The terminal shall process the transaction online until completion with Approval.

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2CA.124.03 Application Version Number assigned by the payment system

Test No: **2CA.124.03** - *Revision* 43j - *Number of sub cases:* 6

Objective: To ensure that the terminal maintains the Application Dependent Data for each supported AID(s) assigned by the payment system.

Reference EMV 4.3: Book 4 - Section 10.2 - Application Dependent Data

Terminal Configuration: [Offline/Online capable] supported OR ([Online Only AND Normal Default Action processing]) supported

Terminal Configuration (New format): [Offline/Online Capable] OR ([Online Only] AND [Normal Default Action Code processing])

Conditions:

- Terminal is configured <u>at the same time</u> with 'NbAIDsForTest' different AIDs and each of them have a set of different data according to table 7 (excluding the following data elements managed at the Terminal Level: Transaction Currency Code, Transaction Currency Exponent, Transaction Reference Currency Code and Transaction Reference Currency Exponent) of section 10.2, Book 4 and for each AID the data values are different.
    - The number of AIDs 'NbAIDsForTest' configured in the Terminal depends on the response of the ICS question: 'How Many AIDs can be supported'= NbMaxAIDs, where NbAIDsForTest=MIN(11, NbMaxAIDs).
- For AID 1: Terminal Action Code – Default, Denial and Online are set to 0
- For AID 2: Terminal Action Code – Online, and Denial are set to '0' Terminal Action Code – Default are set to 20 00 00 00 00.
- For AID 3: Terminal Action Code – Online, and Denial are set to '0' Terminal Action Code – Default are set to 00 80 00 00 00.
- For AID 4: Terminal Action Code – Online, and Denial are set to '0' Terminal Action Code – Default are set to 00 00 40 00 00.
- For AID 5: Terminal Action Code – Online, and Denial are set to '0' Terminal Action Code – Default are set to 00 40 00 00 00.
- For AID 6: Terminal Action Code – Online, and Denial are set to '0' Terminal Action Code – Default are set to 00 20 00 00 00.
- For AID 7 Terminal Action Code – Default, and Denial are set to '0' Terminal Action Code – Online are set to 00 40 00 00 00.
- For AID 8 Terminal Action Code – Default, and Denial are set to '0' Terminal Action Code – Online are set to 0.
- For AID 9 Terminal Action Code – Default, and Denial are set to '0' Terminal Action Code – Online are set to 00 80 00 00 00.
- For AID 10 Terminal Action Code – Default, and Denial are set to '0' Terminal Action Code – Online are set to 00 60 00 00 00.

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2CA.124.03 Application Version Number assigned by the payment system

- For AID 11: Terminal Action Code – Default, Denial and Online are set to 0
- Terminal is unable to go online
- The below subcases are run depending on the number of AID supported, if less than 11 AIDs are supported, the corresponding number of subcases are run
- Case 01: AID 1 is selected
- Case 02: AID 2 is selected and CVM List not present in Card and AIP indicates cardholder verification is supported.
- Case 03: AID 3 is selected and ICC Application Version Number does not match Terminal Application Version Number.
- Case 04: AID 4 is selected and CVM list contain only one rules which is an 'Unrecognised CVM'
- Case 05: AID 5 is selected and Application is expired in Card.
- Case 06: AID 6 is selected and Application is not yet effective in Card.
- Case 07: AID 7 is selected, and Application is expired in Card
- Case 08: AID 08 is selected.
- Case 09: AID 9 is selected and ICC Application Version Number matches Terminal Application Version Number.
- Case 10: AID 10 is selected and Application is effective and not expired in Card.
- Case 11: AID 11 is selected

Action: NA

Procedure: Application in LT is selected and transaction is processed with LT.

Pass Criteria:

- Case 01, 08, 09, 10 and 11: The terminal shall process the transaction until completion with Approval.
- Case 02 to 07: The terminal shall process the transaction until completion with Decline at Second Generate AC.

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2CA.130.00 Tokenisation Data in INTERNAL AUTHENTICATE

Test No: **2CA.130.00** - *Revision* 43j - *Number of sub cases:* 4

Objective: To ensure that Tokenisation Data Objects returned in INTERNAL AUTHENTICATE response are understood by the Kernel.

Reference EMV 4.3: Book Bulletins - Section SB 178 - Bulletin n°178, Tokenisation Data Objects – Payment Account Reference (PAR) (April 2016)

Book Bulletins - Section SB 197 - Bulletin n°197, Tokenisation Data Objects – Token Requestor ID and Last 4 Digits of PAN (May 2017)

Terminal Configuration: [DDA] supported

Terminal Configuration (New format): [DDA]

Conditions:

- CDOL1 requests the PAR Data Object (tag 9F24) and the Token Requestor ID Data Object (tag 9F19) and the Last 4 Digits of PAN Data Object (tag 9F25).
- Case 01: Response to INTERNAL AUTHENTICATE contains PAR Data Object (tag 9F24) and shall be encoded in template 77
- Case 02: Response to INTERNAL AUTHENTICATE contains Token Requestor ID Data Object (tag 9F19) and shall be encoded in template 77
- Case 03: Response to INTERNAL AUTHENTICATE contains PAR Data Object (tag 9F24) and Token Requestor ID Data Object (tag 9F19) and shall be encoded in template 77
- Case 04: Response to INTERNAL AUTHENTICATE contains Last 4 Digits of PAN Data Object (tag 9F25) and shall be encoded in template 77

Action: NA

Procedure: Application in LT is selected and transaction is performed online with LT.

Pass Criteria:

- The terminal shall accept the card and process the transaction until completion, by requesting a TC at First or Second GENERATE AC.
- Case 01: The PAR Data Object (tag 9F24) received at 1st GENERATE AC shall have same value as at INTERNAL AUTHENTICATE response. Token Requestor ID Data Object (tag 9F19) and Last 4 Digits of PAN Data Object (tag 9F25) shall be set to ’00..00’.

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2CA.130.00 Tokenisation Data in INTERNAL AUTHENTICATE

- Case 02: The Token Requestor ID Data Object (tag 9F19) received at 1st GENERATE AC shall have same value as at INTERNAL AUTHENTICATE response. PAR Data Object (tag 9F24) and Last 4 Digits of PAN Data Object (tag 9F25) shall be set to ’00..00’.
- Case 03: The PAR Data Object (tag 9F24) and Token Requestor ID Data Object (tag 9F19) received at 1st GENERATE AC shall have same value as at INTERNAL AUTHENTICATE response. and Last 4 Digits of PAN Data Object (tag 9F25) shall be set to ’00..00’.
- Case 04: The Last 4 Digits of PAN Data Object (tag 9F25) received at 1st GENERATE AC shall have same value as received at INTERNAL AUTHENTICATE. Token Requestor ID Data Object (tag 9F19) and PAR Data Object (tag 9F24) shall be set to ’00..00’.
- The data output from the Kernel shall contain the Tokenisation Data Object(s) with same value as above pass criteria (if data present in the ICC, else not verified). Data Ouput presences and values shall be verified by the Test Tool at the end of the transaction.

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2CB.024.00 Selection using a list of AIDs

Test No: **2CB.024.00** - *Revision* 43j - *Number of sub cases:* 0

Objective: To ensure that the terminal is able to manage a selection using a list of AIDs. To ensure that if the terminal uses the list of AID method, it issues the first SELECT command using the first AID in the list.

Reference EMV 4.3: Book 1 - Section 12.3.3 - Using a List of AIDs

Terminal Configuration: NA

Terminal Configuration (New format): [NA]

Conditions:

- Terminal is configured at the same time with 'NbAIDsForTest' different AIDs and each of them have a different FCI response.
  - The number of AIDs 'NbAIDsForTest' configured in the Terminal depends on the response of the ICS question: 'How Many AIDs can be supported'= NbMaxAIDs, where NbAIDsForTest=MIN(11, NbMaxAIDs).
- LT does not contain PSE
- LT is set up to have the same number x of ADF as the Terminal
- LT is set up to have the same ADF values as the Terminal
- For each ADF, FCI of ADF contains all optional data objects: Application Priority Indicator ('87'), PDOL ('9F38'), Language Preference ('5F 2D'), Issuer Code Table Index ('9F 11'), Application Preferred Name ('9F 12') and FCI issuer Discretionary Data ('BF 0C') containing: '9F 4D' '5F54' Bank Identifier Code (BIC), '5F53' International Bank Account Number (IBAN), '5F55' Issuer Country Code (alpha 2), '5F56' Issuer Country Code (alpha 3), '42' Issuer Identifier Number and the EMV reserved tag '9F30' with a length such that FCI template reach its maximum length.

Action: NA

Procedure: Application Selection using list of AID is performed.

Pass Criteria:

- The terminal shall process the transaction until completion, by requesting a TC.
- LT shall receive 'NbAIDsForTest' SELECT 'AID' command.

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2CB.031.00 Application Selection Indicator: One Occurrence Permitted

Test No: **2CB.031.00** - *Revision* 43j - *Number of sub cases:* 0

Objective: To ensure that if the DFNAME in the ICC is longer than that of the AID in the terminal, but they are identical up to and including the last character in the terminal AID, the terminal shall check the Application Selection indicator.
If the indicator says only one occurrence is allowed the terminal does not add the AID to the candidate list and issue another SELECT command using the next AID.

Reference EMV 4.3: Book 1 - Section 12.3.3 - Using a List of AIDs

Terminal Configuration: NA

Terminal Configuration (New format): [NA]

Conditions:

- LT does not contain PSE
- DF Name 1 contained in FCI returned by the LT for first AID (AID1) selection is longer than that of terminal AID1 used for selection, but they are identical up to and including the last character in the terminal AID1.
- Terminal Application Selection Indicator only allows one occurrence for selected AID.
- LT shall contain another DF Name 2 exactly matching the another terminal AID2.

Action: NA

Procedure: Application Selection using list of AIDs is performed.

Pass Criteria:

- The terminal shall process the transaction until completion, by requesting a TC.
- LT shall receive SELECT Command with AID name that exactly matches the ADF name from the LT.
- LT shall not receive SELECT Command with the longer DF Name 1
- After SELECT AID1 command, the LT shall receive an SELECT 'AID' command but with the AID2 name.
- The terminal will not store the AID1 as a selectable application.

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2CB.031.09 DF name longer than AID & multiple occurrence

Test No: **2CB.031.09** - *Revision* 43j - *Number of sub cases:* 0

Objective: To ensure that if the DF name in the ICC is longer than that of the AID in the terminal but are identical up to and including the last character in the terminal AID, the terminal shall check the Application Selection indicator.
If the Application Selection Indicator indicates that multiple occurrence are permitted and if the application is not blocked, the terminal adds the AID to the candidate list and repeats the SELECT command using the same command data as before, but changes P2 in the command to '02'.

Reference EMV 4.3: Book 1 - Section 12.3.3 - Using a List of AIDs

Terminal Configuration: NA

Terminal Configuration (New format): [NA]

Conditions:

- Terminal Application Selection Indicator allows multiple occurrences for selected AID.
- LT does not contain PSE
- LT contain the ADF's corresponding to the AID's supported by the Terminal
- LT returns 20 extended AIDs related to the AID 'A0000000031010'. For each occurence:
  - The DF Name returned in FCI by the LT are longer than terminal AID A0000000031010 used for selection, but they are identical up to and including the last character in the terminal AID (So A00000000310100001, A00000000310100002, etc).
  - For each ADF, FCI of ADF contains all optional data objects: Application Priority Indicator ('87'), PDOL ('9F38'), Language Preference ('5F 2D'), Issuer Code Table Index ('9F 11'), Application Preferred Name ('9F 12') and FCI issuer Discretionary Data ('BF 0C') containing: '9F 4D' '5F54' Bank Identifier Code (BIC), '5F53' International Bank Account Number (IBAN), '5F55' Issuer Country Code (alpha 2), '5F56' Issuer Country Code (alpha 3), '42' Issuer Identifier Number and the EMV reserved tag '9F30' with a length such that FCI template reach its maximum length.
- Application Priority Indicator of the ADFs are different and in such a way that the ADF with highest priority is the one returned on the last SELECT NEXT command (so on DF with value A00000000310100020).
- Case 01: Terminal supports three AID's:
  - A0000000031010

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2CB.031.09 DF name longer than AID & multiple occurrence

- A0000000041010
- A0000000250000

- Case 02: Terminal supports only one AID's:
- A0000000031010

Action: NA

Procedure: Application Selection using list of AIDs is performed.

Pass Criteria:

- The terminal shall process the transaction until completion, by requesting a TC
- LT shall receive twenty (20) SELECT 'AIDs' with P2 option set to 'next'.
- The terminal shall select the AID (A00000000310100020) and process the transaction using that AID.

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2CB.031.10 DF name longer than AID & multiple occurrence (2)

Test No: **2CB.031.10** - *Revision* 43j - *Number of sub cases:* 0

Objective: To ensure that if the DF name in the ICC is longer than that of the AID in the terminal but are identical up to and including the last character in the terminal AID, the terminal shall check the Application Selection indicator.
If the Application Selection Indicator indicates that multiple occurrence are permitted and if the application is not blocked, the terminal adds the AID to the candidate list and repeats the SELECT command using the same command data as before, but changes P2 in the command to '02'.

Reference EMV 4.3: Book 1 - Section 12.3.3 - Using a List of AIDs

Terminal Configuration: NA

Terminal Configuration (New format): [NA]

Conditions:

- Terminal supports three AID's:
  - A0000000031010
  - A0000000041010
  - A0000000250000
- Terminal Application Selection Indicator allows multiple occurrences for selected AID.
- LT does not contain PSE
- LT contain the ADF's corresponding to the AID's supported by the Terminal
- LT returns 5 extended AIDs related to the AID 'A0000000031010'. For each occurence:
  - The DF Name returned in FCI by the LT are longer than terminal AID A0000000031010 used for selection, but they are identical up to and including the last character in the terminal AID (So A00000000310100001, A00000000310100002, etc).
  - For each ADF, FCI of ADF contains all optional data objects: Application Priority Indicator ('87'), PDOL ('9F38'), Language Preference ('5F 2D'), Issuer Code Table Index ('9F 11'), Application Preferred Name ('9F 12') and FCI issuer Discretionary Data ('BF 0C') containing: '9F 4D' '5F54' Bank Identifier Code (BIC), '5F53' International Bank Account Number (IBAN), '5F55' Issuer Country Code (alpha 2), '5F56' Issuer Country Code (alpha 3), '42' Issuer Identifier Number and the EMV reserved tag '9F30' with a length such that FCI template reach its maximum length.
- LT returns 11 extended AIDs related to the AID 'A0000000041010'. For each occurence:

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2CB.031.10 DF name longer than AID & multiple occurrence (2)

- The DF Name returned in FCI by the LT are longer than terminal AID A0000000041010 used for selection, but they are identical up to and including the last character in the terminal AID (So A00000000410100001, A00000000410100002, etc).
- For each ADF, FCI of ADF contains all optional data objects: Application Priority Indicator (‘87’), PDOL (‘9F38’), Language Preference (‘5F 2D’), Issuer Code Table Index (‘9F 11’), Application Preferred Name (‘9F 12’) and FCI issuer Discretionary Data (‘BF 0C’) containing: ‘9F 4D’ '5F54' Bank Identifier Code (BIC), '5F53' International Bank Account Number (IBAN), '5F55' Issuer Country Code (alpha 2), '5F56' Issuer Country Code (alpha 3), '42' Issuer Identifier Number and the EMV reserved tag ‘9F30’ with a length such that FCI template reach its maximum length.

- LT returns 4 extended AIDs related to the AID ‘A0000000250000’. For each occurence:
  - The DF Name returned in FCI by the LT are longer than terminal AID A0000000250000 used for selection, but they are identical up to and including the last character in the terminal AID (So A00000002500000001, A00000002500000002, etc).
  - For each ADF, FCI of ADF contains all optional data objects: Application Priority Indicator (‘87’), PDOL (‘9F38’), Language Preference (‘5F 2D’), Issuer Code Table Index (‘9F 11’), Application Preferred Name (‘9F 12’) and FCI issuer Discretionary Data (‘BF 0C’) containing: ‘9F 4D’ '5F54' Bank Identifier Code (BIC), '5F53' International Bank Account Number (IBAN), '5F55' Issuer Country Code (alpha 2), '5F56' Issuer Country Code (alpha 3), '42' Issuer Identifier Number and the EMV reserved tag ‘9F30’ with a length such that FCI template reach its maximum length.
- Application Priority Indicator of the ADFs are different and in such a way that the ADF with highest priority is the one returned on the last SELECT NEXT command (so on DF with value A00000002500000004).

Action: NA

Procedure: Application Selection using list of AIDs is performed.

Pass Criteria:

- The terminal shall process the transaction until completion, by requesting a TC
- LT shall receive twenty (20) SELECT 'AIDs' with P2 option set to 'next'.

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2CB.031.10 DF name longer than AID & multiple occurrence (2)

- The terminal shall select the AID (A00000002500000004) and process the transaction using that AID.

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2CC.071.00 Dynamic Signature generation

Test No: **2CC.071.00** - *Revision* 43j - *Number of sub cases:* 0

Objective: To ensure that the terminal supports a valid DDOL.
To ensure that if Dynamic Data Authentication is supported, the terminal is able to issue an INTERNAL AUTHENTICATE command including data elements specified in DDOL.

Reference EMV 4.3: Book 2 - Section 6.5.1 - Dynamic Signature Generation
Book 3 - Section 5.4 - Rules for Using a Data Object List (DOL)

Terminal Configuration: [DDA] supported

Terminal Configuration (New format): [DDA]

Conditions:
- *AIP of LT indicates Dynamic Data Authentication is supported (AIP byte 1 bit 6 = '1').*
- *DDOL is present in LT.*
- *Dynamic signature computed by LT is correct.*

Action: NA

Procedure: Application in LT is selected and transaction is processed with LT (in particular Dynamic Data Authentication).

Pass Criteria:
- *The terminal shall process the transaction until completion, by requesting a TC or an AAC.*
- *TVR byte 1, bit 4 = '0' (ie Offline Dynamic Data Authentication succeeded) received at 1st GENERATE AC.*
- *TVR byte 1, bit 3 = '0' (ie Combined Data Authentication not used) received at 1st GENERATE AC.*
- *TVR byte 1, bit 7 = '0' (ie Static Data Authentication not used) received at 1st GENERATE AC.*
- *The LT shall receive an INTERNAL AUTHENTICATE command with concatenation of data in DDOL according to rule described in Book 3 Section 5.4.*
- *TSI byte 1, bit 8 = '1' (ie Offline Data Authentication was performed) received at 1st GENERATE AC.*

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2CC.146.00 Source of Unpredictable Number

Test No: **2CC.146.00** - *Revision* 43j - *Number of sub cases:* 0

Objective: To ensure that the terminal generates a different random number from one transaction to another at Get Processing Options.

Reference EMV 4.3: Book 2 - Section 6.5.6 – Unpredictable Number

Terminal Configuration:

Terminal Configuration (New format): [NA]

Conditions:

- PDOL in LT include Unpredictable Number generated by the terminal (tag 9F 37).

Action: NA

Procedure: Three transactions are performed. The unpredictable number values generated by the terminal will be compared.

Pass Criteria:

- The terminal shall process the transaction until completion, by requesting a TC.
- Tag 9F37 value populated in the GPO command shall be not zeros and shall be different for each transaction

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2CI.021.00 Data retrievable by GET PROCESSING OPTIONS command

Test No: **2CI.021.00** - *Revision* 43j - *Number of sub cases:* 0

Objective: To ensure that the terminal is able to retrieve and understand both AFL and AIP, using the GET PROCESSING OPTIONS command.

Reference EMV 4.3: Book 3 - Section 7.4 - Data Retrievable by GET PROCESSING OPTIONS

Terminal Configuration: NA

Terminal Configuration (New format): [NA]

Conditions:

- AFL and AIP are present in the LT.
- AIP of LT indicates Static Data Authentication is supported (AIP byte 1 bit 7 = ‘1’).
- AIP of LT indicates Dynamic Data Authentication is supported (AIP byte 1 bit 6 = ‘1’).
- AIP of LT indicates Cardholder Verification is supported (AIP byte 1 bit 5 = ‘1’).
- AIP of LT indicates Issuer Authentication is supported (AIP byte 1 bit 3 = 1).
- AIP of LT indicates Combined Authentication is supported (AIP byte 1 bit 1 = ‘1’).

Action: NA

Procedure: Application in LT is selected and transaction is processed with LT.

Pass Criteria:

- The terminal shall process the transaction until completion, by requesting a TC.
- LT shall receive a GET PROCESSING OPTIONS command.
- LT shall receive a sequence of READ RECORD according to AFL.

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2CJ.002.00 All bits in TSI and TVR are set to 0b

Test No: **2CJ.002.00** - *Revision* 43j - *Number of sub cases:* 0

Objective: To ensure that all TVR and TSI bits are set to 0b when the terminal starts the transaction.

Reference EMV 4.3: Book 3 - Section 10.1 - Initiate Application Processing

Terminal Configuration: NA

Terminal Configuration (New format): [NA]

Conditions:

- A first transaction is performed raising the TVR bits: Byte 2 bit 8 to bit 5 (ICC have different application version, Expired Application, Application not yet effective, Requested service not allowed for card product), Byte 3 bit 8 to 7 (Cardholder verification not successful, Unrecognised CVM) and raising the TSI bits: Byte 1 bit 7 to 6 (Cardholder verification was performed, Card risk management was performed).
- A second transaction is performed where the PDOL requests TSI and TVR.

Action: NA

Procedure: Application in LT is selected and transaction is processed with LT.

Pass Criteria:

- TVR byte 2 bit 8 to bit 5, byte 3 bit 8 to 7 = ‘1’ received at 1st GENERATE AC in the first transaction.
- TSI byte 1, bit 7 = '1' (ie Cardholder verification was performed) received at 1st GENERATE AC in the first transaction.
- TSI returned by terminal with GET PROCESSING OPTIONS shall be set to 0b in the second transaction.
- TVR returned by terminal with GET PROCESSING OPTIONS shall be set to 0b in the second transaction.

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2CJ.002.01 All bits in TSI and TVR are set to 0b (2)

Test No: **2CJ.002.01** - *Revision* 43j - *Number of sub cases:* 2

Objective: To ensure that all TVR and TSI bits are set to 0b when the terminal starts the transaction.

Reference EMV 4.3: Book 3 - Section 10.1 - Initiate Application Processing

Terminal Configuration: [SDA] supported

Terminal Configuration (New format): [SDA]

Conditions:

- Case 01:
  - A first transaction is performed raising the TVR bits: Byte 1 bit 7 (SDA failed), Byte 2 bit 8 to 5 (ICC have different application version, Expired Application, Application not yet effective, Requested service not allowed for card product), Byte 3 bit 8- to 7 (Cardholder verification not successful, Unrecognised CVM) and raising the TSI bits: Byte 1 bit 8 to 6 (Offline data authentication was performed, Cardholder verification was performed, Card risk management was performed).
  - A second transaction is performed where the PDOL requests TSI and TVR.
- Case 02:
  - A first transaction is performed raising the TVR bits: Byte 1 bit 8 (Offline Data Authentication not performed)
  - A second transaction is performed where the PDOL requests TSI and TVR.

Action: NA

Procedure: Application in LT is selected and transaction is processed with LT.

Pass Criteria:

- Case 01: TVR byte 1 bit 7, byte 2 bit 8 to 5, byte 3 bit 8 to 7 = ‘1’ received at 1st GENERATE AC in the first transaction.
- Case 01: TSI byte 1, bit 8 to 7 = '1' received at 1st GENERATE AC in the first transaction.
- Case 02: TVR byte 1 bit 8 = ‘1’ received at 1st GENERATE AC in the first transaction.
- TSI returned by terminal with GET PROCESSING OPTIONS shall be set to 0b in the second transaction.
- TVR returned by terminal with GET PROCESSING OPTIONS shall be set to 0b in the second transaction.

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2CJ.002.02 All bits in TSI and TVR are set to 0b (3)

Test No: **2CJ.002.02** - *Revision* 43j - *Number of sub cases:* 0

Objective: To ensure that all TVR and TSI bits are set to 0b when the terminal starts the transaction.

Reference EMV 4.3: Book 3 - Section 10.1 - Initiate Application Processing

Terminal Configuration: [Exception File] supported

Terminal Configuration (New format): [Exception File]

Conditions:

- A first transaction is performed raising the TVR bits: Byte 1 bit 5 (Card appears in exception file), Byte 2 bit 8 to 5 (ICC have different application version, Expired Application, Application not yet effective, Requested service not allowed for card product), Byte 3 bit 8 to 7 (Cardholder verification not successful, Unrecognised CVM) and raising the TSI bits: Byte 1 bit 7 to 6 (Cardholder verification was performed, Card risk management was performed).
- A second transaction is performed where the PDOL requests TSI and TVR.

Action: NA

Procedure: Application in LT is selected and transaction is processed with LT.

Pass Criteria:

- TVR byte 1 bit 5, byte 2 bit 8 to 5, byte 3 bit 8 to 7 = ‘1’ received at 1st GENERATE AC in the first transaction.
- TSI byte 1, bit 7 = '1' received at 1st GENERATE AC in the first transaction.
- TSI returned by terminal with GET PROCESSING OPTIONS shall be set to 0b in the second transaction.
- TVR returned by terminal with GET PROCESSING OPTIONS shall be set to 0b in the second transaction.

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2CJ.002.03 All bits in TSI and TVR are set to 0b (4)

Test No: **2CJ.002.03** - *Revision* 43j - *Number of sub cases:* 2

Objective: To ensure that all TVR and TSI bits are set to 0b when the terminal starts the transaction.

Reference EMV 4.3: Book 3 - Section 10.1 - Initiate Application Processing

Terminal Configuration: [DDA] supported

Terminal Configuration (New format): [DDA]

Conditions:

- Case 01:
  - A first transaction is performed raising the TVR bits: Byte 1 bit 4 (DDA failed), Byte 2 bit 8 to 5 (ICC have different application version, Expired Application, Application not yet effective, Requested service not allowed for card product), Byte 3 bit 8 to 7 (Cardholder verification not successful, Unrecognised CVM) and raising the TSI bits: Byte 1 bit 8 to 6 (Offline data authentication was performed, Cardholder verification was performed, Card risk management was performed).
  - A second transaction is performed where the PDOL requests TSI and TVR.
- Case 02:
  - A first transaction is performed raising the TVR bits: Byte 1 bit 8 (Offline Data Authentication not performed)
  - A second transaction is performed where the PDOL requests TSI and TVR

Action: NA

Procedure: Application in LT is selected and transaction is processed with LT.

Pass Criteria:

- Case 01: TVR byte 1 bit 4, byte 2 bit 8 to 5, byte 3 bit 8 to 7 = ‘1’ received at 1st GENERATE AC in the first transaction.
- Case 01: TSI byte 1, bit 8 to 7 = '1' received at 1st GENERATE AC in the first transaction.
- Case 02: TVR byte 1 bit 8 = ‘1’ received at 1st GENERATE AC in the first transaction.
- TSI returned by terminal with GET PROCESSING OPTIONS shall be set to 0b in the second transaction.
- TVR returned by terminal with GET PROCESSING OPTIONS shall be set to 0b in the second transaction.

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2CJ.002.04 All bits in TSI and TVR are set to 0b (5)

Test No: **2CJ.002.04** - *Revision* 43j - *Number of sub cases:* 2

Objective: To ensure that all TVR and TSI bits are set to 0b when the terminal starts the transaction.

Reference EMV 4.3: Book 3 - Section 10.1 - Initiate Application Processing

Terminal Configuration: [CDA] supported

Terminal Configuration (New format): [CDA]

Conditions:

- Case 01:
  - A first transaction is performed raising the TVR bits: Byte 1 bit 3 (CDA failed), Byte 2 bit 8 to 5 (ICC have different application version, Expired Application, Application not yet effective, Requested service not allowed for card product), Byte 3 bit 8 -to 7 (Cardholder verification not successful, Unrecognised CVM) and raising the TSI bits: Byte 1 bit 8 to 6 (Offline data authentication was performed, Cardholder verification was performed, Card risk management was performed).
  - A second transaction is performed where the PDOL requests TSI and TVR.
- Case 02:
  - A first transaction is performed raising the TVR bits: Byte 1 bit 8 (Offline Data Authentication not performed)
  - A second transaction is performed where the PDOL requests TSI and TVR

Action: NA

Procedure: Application in LT is selected and transaction is processed with LT.

Pass Criteria:

- Case 01: TVR byte 1 bit 3, byte 2 bit 8 to 5, byte 3 bit 8 to 7 = ‘1’ received at 1st GENERATE AC in the first transaction.
- Case 01: TSI byte 1, bit 8 to 7 = '1' received at 1st GENERATE AC in the first transaction.
- Case 02: TVR byte 1 bit 8 = ‘1’ received at 1st GENERATE AC in the first transaction.
- TSI returned by terminal with GET PROCESSING OPTIONS shall be set to 0b in the second transaction.
- TVR returned by terminal with GET PROCESSING OPTIONS shall be set to 0b in the second transaction.

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2CJ.002.05 All bits in TSI and TVR are set to 0b (6)

Test No: **2CJ.002.05** - *Revision* 43j - *Number of sub cases:* 0

Objective: To ensure that all TVR and TSI bits are set to 0b when the terminal starts the transaction.

Reference EMV 4.3: Book 3 - Section 10.1 - Initiate Application Processing

Terminal Configuration: [Offline Plaintext PIN] supported OR [Offline Enciphered PIN] supported

Terminal Configuration (New format): [Offline Plaintext PIN] OR [Offline Enciphered PIN]

Conditions:

- A first transaction is performed raising the TVR bits: Byte 2 bit 8 to 5 (ICC have different application version, Expired Application, Application not yet effective, Requested service not allowed for card product), Byte 3 bit 8,6 (Cardholder verification not successful, PIN Try Limit exceeded) and raising the TSI bits: Byte 1 bit 7 to 6 (Cardholder verification was performed, Card risk management was performed).
- A second transaction is performed where the PDOL requests TSI and TVR.

Action: NA

Procedure: Application in LT is selected and transaction is processed with LT.

Pass Criteria:

- TVR byte 2 bit 8 to 5, byte 3 bit 8, 6 = ‘1’ received at 1st GENERATE AC in the first transaction.
- TSI returned by terminal with GET PROCESSING OPTIONS shall be set to 0b in the second transaction.
- TVR returned by terminal with GET PROCESSING OPTIONS shall be set to 0b in the second transaction.

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2CJ.002.06 All bits in TSI and TVR are set to 0b (7)

Test No: **2CJ.002.06** - *Revision* 43j - *Number of sub cases:* 0

Objective: To ensure that all TVR and TSI bits are set to 0b when the terminal starts the transaction.

Reference EMV 4.3: Book 3 - Section 10.1 - Initiate Application Processing

Terminal Configuration: [Offline Plaintext PIN] not supported AND [Offline Enciphered PIN] not supported

Terminal Configuration (New format): NOT[Offline Plaintext PIN] AND NOT[Offline Enciphered PIN]

Conditions:

- A first transaction is performed raising the TVR bits: Byte 2 bit 8 to 5 (ICC have different application version, Expired Application, Application not yet effective, Requested service not allowed for card product), Byte 3 bit 8, 5 (Cardholder verification not successful, PIN entry required, PIN pad not present or not working) and raising the TSI bits: Byte 1 bit 7 to 6 (Cardholder verification was performed, Card risk management was performed).
- A second transaction is performed where the PDOL requests TSI and TVR.

Action: NA

Procedure: Application in LT is selected and transaction is processed with LT.

Pass Criteria:

- TVR byte 2 bit 8 to 5, byte 3 bit 8, 5 = ‘1’ received at 1st GENERATE AC in the first transaction.
- TSI byte 1, bit 7 = '1' received at 1st GENERATE AC in the first transaction.
- TSI returned by terminal with GET PROCESSING OPTIONS shall be set to 0b in the second transaction.
- TVR returned by terminal with GET PROCESSING OPTIONS shall be set to 0b in the second transaction.

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2CJ.002.07 All bits in TSI and TVR are set to 0b (8)

Test No: **2CJ.002.07** - *Revision* 43j - *Number of sub cases:* 0

Objective: To ensure that all TVR and TSI bits are set to 0b when the terminal starts the transaction.

Reference EMV 4.3: Book 3 - Section 10.1 - Initiate Application Processing

Terminal Configuration: [Online Only] supported OR [Offline/Online capable] supported

Terminal Configuration (New format): [Online Only] OR [Offline/Online Capable]

Conditions:

- A first transaction is performed raising the TVR bits: Byte 2 bit 8 to 5 (ICC have different application version, Expired Application, Application not yet effective, Requested service not allowed for card product), Byte 3 bit 8 (Cardholder verification not successful) and Byte 5 bit 6 to 5 (Script processing failed before final GENERATE AC, Script processing failed after final GENERATE AC) and raising the TSI bits: Byte 1 bit 7 to 6, 3 (Cardholder verification was performed, Card risk management was performed, script processing was performed).
- A second transaction is performed where the PDOL requests TSI and TVR.

Action: NA

Procedure: Application in LT is selected and transaction is processed with LT.

Pass Criteria:

- TVR byte 2 bit 8 to 5, byte 3 bit 8 = ‘1’ received at 1st GENERATE AC in the first transaction.
- TVR byte 5 bit 6 = ‘1’ received at 2nd GENERATE AC in the first transaction.
- TSI byte 1, bit 7 = '1' received at 1st GENERATE AC in the first transaction.
- TSI byte 1, bit 3 = '1' received at 2nd GENERATE AC in the first transaction.
- TSI returned by terminal with GET PROCESSING OPTIONS shall be set to 0b in the second transaction.
- TVR returned by terminal with GET PROCESSING OPTIONS shall be set to 0b in the second transaction.

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2CJ.002.08 All bits in TSI and TVR are set to 0b (9)

Test No: **2CJ.002.08** - *Revision* 43j - *Number of sub cases:* 0

Objective: To ensure that all TVR and TSI bits are set to 0b when the terminal starts the transaction.

Reference EMV 4.3: Book 3 - Section 10.1 - Initiate Application Processing

Terminal Configuration: [Online Only] OR [Offline/Online capable] supported

Terminal Configuration (New format): [Online Only] OR [Offline/Online Capable]

Conditions:

- A first transaction is performed raising the TVR bits: Byte 2 bit 8 to 5 (ICC have different application version, Expired Application, Application not yet effective, Requested service not allowed for card product), Byte 3 bit 8 to-7 (Cardholder verification not successful, Unrecognised CVM) and Byte 5 bit 7 to 5 (Issuer Authentication failed, Script processing failed before final GENERATE AC, Script processing failed after final GENERATE AC) and raising the TSI bits: Byte 1 bit 7 to 5, 3 (Cardholder verification was performed, Card risk management was performed, Issuer Authentication was performed, script processing was performed).
- A second transaction is performed where the PDOL requests TSI and TVR.

Action: NA

Procedure: Application in LT is selected and transaction is processed with LT.

Pass Criteria:

- TVR byte 2 bit 8 to 5, byte 3 bit 8 to 7 = ‘1’ received at 1st GENERATE AC in the first transaction.
- TVR byte 5 bit 7 to 6 = ‘1’ received at 2nd GENERATE AC in the first transaction.
- TSI byte 1, bit 7 = '1' received at 1st GENERATE AC in the first transaction.
- TSI byte 1, bit 5 = '1' received at 2nd GENERATE AC in the first transaction.
- TSI byte 1, bit 3 = '1' received at 2nd GENERATE AC in the first transaction.
- TSI returned by terminal with GET PROCESSING OPTIONS shall be set to 0b in the second transaction.
- TVR returned by terminal with GET PROCESSING OPTIONS shall be set to 0b in the second transaction.

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2CJ.002.09 All bits in TSI and TVR are set to 0b (10)

Test No: **2CJ.002.09** - *Revision* 43j - *Number of sub cases:* 0

Objective: To ensure that all TVR and TSI bits are set to 0b when the terminal starts the transaction.

Reference EMV 4.3: Book 3 - Section 10.1 - Initiate Application Processing

Terminal Configuration: [Floor Limit Checking] supported

Terminal Configuration (New format): [Floor Limit Checking]

Conditions:

- A first transaction is performed raising the TVR bits: Byte 2 bit 8 to 5 (ICC have different application version, Expired Application, Application not yet effective, Requested service not allowed for card product), Byte 3 bit 8 to 7 (Cardholder verification not successful, Unrecognised CVM) Byte 4 bit 8 (transaction exceeds floor limit) and raising the TSI bits: Byte 1 bit 7 to 6, 4 (Cardholder verification was performed, Card risk management was performed, Terminal risk management was performed).
- A second transaction is performed where the PDOL requests TSI and TVR.

Action: NA

Procedure: Application in LT is selected and transaction is processed with LT.

Pass Criteria:

- TVR byte 2 bit 8 to 5, byte 3 bit 8 to 7, byte 4 bit 8 = ‘1’ received at 1st GENERATE AC in the first transaction.
- TSI byte 1, bit 7 = '1' received at 1st GENERATE AC in the first transaction.
- TSI byte 1, bit 4 = '1' received at 1st GENERATE AC in the first transaction.
- TSI returned by terminal with GET PROCESSING OPTIONS shall be set to 0b in the second transaction.
- TVR returned by terminal with GET PROCESSING OPTIONS shall be set to 0b in the second transaction.

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2CJ.002.10 All bits in TSI and TVR are set to 0b (11)

Test No: **2CJ.002.10** - *Revision* 43j - *Number of sub cases:* 0

Objective: To ensure that all TVR and TSI bits are set to 0b when the terminal starts the transaction.

Reference EMV 4.3: Book 3 - Section 10.1 - Initiate Application Processing

Terminal Configuration: [Velocity Checking] supported

Terminal Configuration (New format): [Velocity Checking]

Conditions:

- A first transaction is performed raising the TVR bits: Byte 2 bit 8 to 4 (ICC have different application version, Expired Application, Application not yet effective, Requested service not allowed for card product, new card), Byte 3 bit 8 to 7 (Cardholder verification not successful, Unrecognised CVM) Byte 4 bit 7 to 6 (Upper consecutive offline limit exceeded, Lower consecutive offline limit exceeded) and raising the TSI bits: Byte 1 bit 7 to 6, 4 (Cardholder verification was performed, Card risk management was performed, Terminal risk management was performed).
- A second transaction is performed where the PDOL requests TSI and TVR.

Action: NA

Procedure: Application in LT is selected and transaction is processed with LT.

Pass Criteria:

- TVR byte 2 bit 8 to 4, byte 3 bit 8 to 7, byte 4 bit 7 to 6 = ‘1’ received at 1st GENERATE AC in the first transaction.
- TSI byte 1, bit 7 = '1' received at 1st GENERATE AC in the first transaction.
- TSI byte 1, bit 4 = '1' received at 1st GENERATE AC in the first transaction.
- TSI returned by terminal with GET PROCESSING OPTIONS shall be set to 0b in the second transaction.
- TVR returned by terminal with GET PROCESSING OPTIONS shall be set to 0b in the second transaction.

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2CJ.002.11 All bits in TSI and TVR are set to 0b (12)

Test No: **2CJ.002.11** - *Revision* 43j - *Number of sub cases:* 0

Objective: To ensure that all TVR and TSI bits are set to 0b when the terminal starts the transaction.

Reference EMV 4.3: Book 3 - Section 10.1 - Initiate Application Processing

Terminal Configuration: [Random Transaction Selection] supported

Terminal Configuration (New format): [Random Transaction Selection]

Conditions:

- A first transaction is performed raising the TVR bits: Byte 2 bit 8 to 5 (ICC have different application version, Expired Application, Application not yet effective, Requested service not allowed for card product), Byte 3 bit 8 to 7 (Cardholder verification not successful, Unrecognised CVM) Byte 4 bit 5 (transaction selected randomly for online processing) and raising the TSI bits: Byte 1 bit 7 to 6, 4 (Cardholder verification was performed, Card risk management was performed, Terminal risk management was performed).
- A second transaction is performed where the PDOL requests TSI and TVR.

Action: NA

Procedure: Application in LT is selected and transaction is processed with LT.

Pass Criteria:

- TVR byte 2 bit 8 to 5, byte 3 bit 8 to 7, byte 4 bit 5 = ‘1’ received at 1st GENERATE AC in the first transaction.
- TSI byte 1, bit 7 = '1' received at 1st GENERATE AC in the first transaction.
- TSI byte 1, bit 4 = '1' received at 1st GENERATE AC in the first transaction.
- TSI returned by terminal with GET PROCESSING OPTIONS shall be set to 0b in the second transaction.
- TVR returned by terminal with GET PROCESSING OPTIONS shall be set to 0b in the second transaction.

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2CJ.002.12 All bits in TSI and TVR are set to 0b (13)

Test No: **2CJ.002.12** - *Revision* 43j - *Number of sub cases:* 0

Objective: To ensure that all TVR and TSI bits are set to 0b when the terminal starts the transaction.

Reference EMV 4.3: Book 3 - Section 10.1 - Initiate Application Processing

Terminal Configuration: [Attended] supported AND [Forced Online] supported

Terminal Configuration (New format): [Attended] AND [Forced Online]

Conditions:

- A first transaction is performed raising the TVR bits: Byte 2 bit 8 to 5 (ICC have different application version, Expired Application, Application not yet effective, Requested service not allowed for card product), Byte 3 bit 8 to 7 (Cardholder verification not successful, Unrecognised CVM) Byte 4 bit 4 (Merchant forced transaction online) and raising the TSI bits: Byte 1 bit 7 to 6, 4 (Cardholder verification was performed, Card risk management was performed, Terminal risk management was performed).
- A second transaction is performed where the PDOL requests TSI and TVR.

Action: NA

Procedure: Application in LT is selected and transaction is processed with LT.

Pass Criteria:

- TVR byte 2 bit 8 to 5, byte 3 bit 8 to 7, byte 4 bit 4 = ‘1’ received at 1st GENERATE AC in the first transaction.
- TSI byte 1, bit 7 = '1' received at 1st GENERATE AC in the first transaction.
- TSI byte 1, bit 4 = '1' received at 1st GENERATE AC in the first transaction.
- TSI returned by terminal with GET PROCESSING OPTIONS shall be set to 0b in the second transaction.
- TVR returned by terminal with GET PROCESSING OPTIONS shall be set to 0b in the second transaction.

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2CJ.002.13 All bits in TSI and TVR are set to 0b (14)

Test No: **2CJ.002.13** - *Revision* 43j - *Number of sub cases:* 0

Objective: To ensure that all TVR and TSI bits are set to 0b when the terminal starts the transaction.

Reference EMV 4.3: Book 3 - Section 10.1 - Initiate Application Processing

Terminal Configuration: [Online Enciphered PIN] supported

Terminal Configuration (New format): [Online Enciphered PIN]

Conditions:

- A first transaction is performed raising the TVR bits: Byte 2 bit 8 to 5 (ICC have different application version, Expired Application, Application not yet effective, Requested service not allowed for card product), Byte 3 bit 3 (Online PIN entered) and Byte 5 bit 6 to 5 (Script processing failed before final GENERATE AC, Script processing failed after final GENERATE AC) and raising the TSI bits: Byte 1 bit 7 to 6, 3 (Cardholder verification was performed, Card risk management was performed, script processing was performed).
- A second transaction is performed where the PDOL requests TSI and TVR.

Action: NA

Procedure: Application in LT is selected and transaction is processed with LT.

Pass Criteria:

- TVR byte 2 bit 8 to 5, byte 3 bit 3 = ‘1’ received at 1st GENERATE AC in the first transaction.
- TVR byte 5 bit 6 = ‘1’ received at 2nd GENERATE AC in the first transaction.
- TSI byte 1, bit 7 = '1' received at 1st GENERATE AC in the first transaction.
- TSI byte 1, bit 4 = '1' received at 1st GENERATE AC in the first transaction.
- TSI returned by terminal with GET PROCESSING OPTIONS shall be set to 0b in the second transaction.
- TVR returned by terminal with GET PROCESSING OPTIONS shall be set to 0b in the second transaction.

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2CJ.002.14 All bits in TSI and TVR are set to 0b (15)

Test No: **2CJ.002.14** - *Revision* 43j - *Number of sub cases:* 0

Objective: To ensure that all TVR and TSI bits are set to 0b when the terminal starts the transaction.

Reference EMV 4.3: Book 3 - Section 10.1 - Initiate Application Processing

Terminal Configuration: [Default TDOL] supported

Terminal Configuration (New format): [Default TDOL]

Conditions:

- A first transaction is performed raising the TVR bit: Byte 5 bit 8 (Default TDOL used)
- A second transaction is performed where the PDOL requests TSI and TVR

Action: NA

Procedure: Application in LT is selected and transaction is processed with LT.

Pass Criteria:

- TVR byte 5 bit 8 = ‘1’ received at 1st GENERATE AC in the first transaction.
- TSI returned by terminal with GET PROCESSING OPTIONS shall be set to 0b in the second transaction.
- TVR returned by terminal with GET PROCESSING OPTIONS shall be set to 0b in the second transaction.

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2CJ.035.00 Processing Restrictions function execution

Test No: **2CJ.035.00** - *Revision* 43j - *Number of sub cases:* 0

Objective: To ensure that the terminal performs processing restrictions functions at some time after Read Application Data and before completion of the terminal action analysis process.

Reference EMV 4.3: Book 3 - Section 10.4 - Processing Restrictions

Terminal Configuration: [Cash] OR [cash back] OR [Good] OR [Services] supported

Terminal Configuration (New format): [Cash] OR [Cashback] OR [Goods] OR [Services]

Conditions:

- Application Version Number is different in LT and Terminal.
- Issuer Country Code matches Terminal Country Code.
- AUC is present in LT with value ‘5740’
  - 'Valid for domestic cash transactions' bit is set to 0b
  - 'Valid for internationl cash transactions' bit is set to 1b
  - 'Valid for domestic goods’ bit is set to 0b.
  - 'Valid for international goods’ bit is set to 1b
  - 'Valid for domestic services’ bit is set to 0b
  - 'Valid for international services’ bit is set to 1b
  - 'Valid at ATMs' bit is set to 1b.
  - 'Valid at terminals other than ATMs' bit is set to 1b
  - 'Domestic cashback allowed' bit is set to 0b.
  - 'International cashback allowed' bit is set to 1b.
- Application Expiration Date in the LT has passed.

Action: NA

Procedure: Application in LT is selected and transaction is processed with LT.

Pass Criteria:

- The terminal shall process the transaction until completion.
- TVR byte 2, bit 8 = '1' (ie ICC and Terminal have different application versions) received at 1st GENERATE AC.
- TVR byte 2, bit 5 = '1' (ie Requested service not allowed for card product) received at 1st GENERATE AC.
- TVR byte 2, bit 7 = '1' (ie Expired application) received at 1st GENERATE AC.

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2CJ.041.00 AUC is present in the ICC and Transaction is conducted at an ATM

Test No: **2CJ.041.00** - *Revision* 43j - *Number of sub cases:* 0

Objective: To ensure that if the terminal is an ATM the AUC is present in the card, and the 'Valid at ATMs' bit is not set to 1b in the AUC, the terminal sets the 'Requested service not allowed for card product' bit in the TVR to 1b.

Reference EMV 4.3: Book 3 - Section 10.4.2 - Application Usage Control

Terminal Configuration: [ATM] supported

Terminal Configuration (New format): [ATM]

Conditions:

- AUC is present in LT with value ‘FDC0’
  - 'Valid for domestic cash transactions' bit is set to 1b
  - 'Valid for internationl cash transactions' bit is set to 1b
  - 'Valid for domestic goods’ bit is set to 1b.
  - 'Valid for international goods’ bit is set to 1b
  - 'Valid for domestic services’ bit is set to 1b
  - 'Valid for international services’ bit is set to 1b
  - 'Valid at ATMs' bit is set to 0b.
  - 'Valid at terminals other than ATMs' bit is set to 1b
  - 'Domestic cashback allowed' bit is set to 1b.
  - 'International cashback allowed' bit is set to 1b.

Action: NA

Procedure: Application in LT is selected and transaction is processed with LT.

Pass Criteria:

- The terminal shall process the transaction until completion, by requesting a TC.
- TVR byte 2, bit 5 = '1' (ie Requested service not allowed for card product) received at 1st GENERATE AC.

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2CJ.042.00 AUC is present in the ICC and Transaction is conducted at an ATM (implied)

Test No: **2CJ.042.00** - *Revision* 43j - *Number of sub cases:* 0

Objective: To ensure that if the terminal is an ATM the AUC is present in the card, and the 'Valid at ATMs' bit is set to 1b in the AUC, the terminal does not set the 'Requested service not allowed for card product' bit in the TVR to 1b.

Reference EMV 4.3: Book 3 - Section 10.4.2 - Application Usage Control

Terminal Configuration: [ATM] supported

Terminal Configuration (New format): [ATM]

Conditions:

- AUC is present in LT with value ‘FFC0’
  - 'Valid for domestic cash transactions' bit is set to 1b
  - 'Valid for internationl cash transactions' bit is set to 1b
  - 'Valid for domestic goods’ bit is set to 1b.
  - 'Valid for international goods’ bit is set to 1b
  - 'Valid for domestic services’ bit is set to 1b
  - 'Valid for international services’ bit is set to 1b
  - 'Valid at ATMs' bit is set to 1b.
  - 'Valid at terminals other than ATMs' bit is set to 1b
  - 'Domestic cashback allowed' bit is set to 1b.
  - 'International cashback allowed' bit is set to 1b.
-

Action: NA

Procedure: Application in LT is selected and transaction is processed with LT.

Pass Criteria:

- The terminal shall process the transaction until completion, by requesting a TC.
- TVR byte 2, bit 5 = '0' (ie Requested service allowed for card product) received at 1st GENERATE AC.

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2CJ.043.00 AUC is present in the ICC and Transaction is not conducted at an ATM

Test No: **2CJ.043.00** - *Revision* 43j - *Number of sub cases:* 0

Objective: To ensure that if the terminal is not an ATM the AUC is present in the card, and the 'Valid at terminals other than ATMs ' bit is not set to 1b in the AUC, the terminal sets the 'Requested service not allowed for card product' bit in the TVR to 1b.

Reference EMV 4.3: Book 3 - Section 10.4.2 - Application Usage Control

Terminal Configuration: [ATM] not supported

Terminal Configuration (New format): NOT[ATM]

Conditions:

- AUC is present in LT with value ‘FEC0’
  - 'Valid for domestic cash transactions' bit is set to 1b
  - 'Valid for internationl cash transactions' bit is set to 1b
  - 'Valid for domestic goods’ bit is set to 1b.
  - 'Valid for international goods’ bit is set to 1b
  - 'Valid for domestic services’ bit is set to 1b
  - 'Valid for international services’ bit is set to 1b
  - 'Valid at ATMs' bit is set to 1b.
  - 'Valid at terminals other than ATMs' bit is set to 0b
  - 'Domestic cashback allowed' bit is set to 1b.
  - 'International cashback allowed' bit is set to 1b.

Action: NA

Procedure: Application in LT is selected and transaction is processed with LT.

Pass Criteria:

- The terminal shall process the transaction until completion, by requesting a TC.
- TVR byte 2, bit 5 = '1' (ie Requested service not allowed for card product) received at 1st GENERATE AC.

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2CJ.044.00 AUC is present in the ICC and Transaction is not conducted at an ATM (implied)

Test No: **2CJ.044.00** - *Revision* 43j - *Number of sub cases:* 0

Objective: To ensure that if the terminal is other than ATM the AUC is present in the card, and the 'Valid at terminals other than ATMs ' bit is set to 1b in the AUC, the terminal does not set the 'Requested service not allowed for card product' bit in the TVR to 1b.

Reference EMV 4.3: Book 3 - Section 10.4.2 - Application Usage Control

Terminal Configuration: [ATM] not supported

Terminal Configuration (New format): NOT[ATM]

Conditions:

- AUC is present in LT with value ‘FFC0’
  - 'Valid for domestic cash transactions' bit is set to 1b
  - 'Valid for internationl cash transactions' bit is set to 1b
  - 'Valid for domestic goods’ bit is set to 1b.
  - 'Valid for international goods’ bit is set to 1b
  - 'Valid for domestic services’ bit is set to 1b
  - 'Valid for international services’ bit is set to 1b
  - 'Valid at ATMs' bit is set to 1b.
  - 'Valid at terminals other than ATMs' bit is set to 1b
  - 'Domestic cashback allowed' bit is set to 1b.
  - 'International cashback allowed' bit is set to 1b.
-

Action: NA

Procedure: Application in LT is selected and transaction is processed with LT.

Pass Criteria:

- The terminal shall process the transaction until completion, by requesting a TC or an AAC.
- TVR byte 2, bit 5 = '0' (ie Requested service allowed for card product) received at 1st GENERATE AC.

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2CJ.045.00 Transaction type is a Cash transaction, and Issuer Country Code = Terminal Country Code

Test No: **2CJ.045.00** - *Revision* 43j - *Number of sub cases:* 0

Objective: To ensure that if the Terminal Country Code matches Issuer Country Code, Transaction Type indicates a cash transaction, the AUC is present in the card, and the 'Valid for domestic cash transactions' bit is not set to 1b in the AUC, the terminal sets the 'Requested service not allowed for card product' bit in the TVR to 1b.

Reference EMV 4.3: Book 3 - Section 10.4.2 - Application Usage Control

Terminal Configuration: [Cash] supported

Terminal Configuration (New format): [Cash]

Conditions:

- Transaction is a cash transaction.
- AUC is present in LT with value ‘7FC0’
  - 'Valid for domestic cash transactions' bit is set to 0b
  - 'Valid for internationl cash transactions' bit is set to 1b
  - 'Valid for domestic goods’ bit is set to 1b.
  - 'Valid for international goods’ bit is set to 1b
  - 'Valid for domestic services’ bit is set to 1b
  - 'Valid for international services’ bit is set to 1b
  - 'Valid at ATMs' bit is set to 1b.
  - 'Valid at terminals other than ATMs' bit is set to 1b
  - 'Domestic cashback allowed' bit is set to 1b.
  - 'International cashback allowed' bit is set to 1b.
- Issuer Country Code matches Terminal Country Code.

'Valid for domestic cash transactions' bit is not set to 1b in the AUC.

Action: NA

Procedure: Application in LT is selected and transaction is processed with LT.

Pass Criteria:

- The terminal shall process the transaction until completion, by requesting a TC.
- TVR byte 2, bit 5 = '1' (ie Requested service not allowed for card product) received at 1st GENERATE AC.
- Transaction Type shall indicate a cash transaction.

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2CJ.046.00 Transaction type is a Cash transaction, and Issuer Country Code = Terminal Country Code (implied)

Test No: **2CJ.046.00** - *Revision* 43j - *Number of sub cases:* 0

Objective: To ensure that if the Terminal Country Code matches Issuer Country Code, Transaction Type indicates a cash transaction, the AUC is present in the card, and the 'Valid for domestic cash transactions' bit is set to 1b in the AUC, the terminal does not set the 'Requested service not allowed for card product' bit in the TVR to 1b.

Reference EMV 4.3: Book 3 - Section 10.4.2 - Application Usage Control

Terminal Configuration: [Cash] supported

Terminal Configuration (New format): [Cash]

Conditions:

- Transaction is a cash transaction.
- Issuer Country Code matches Terminal Country Code.
- AUC is present in LT with value ‘FFC0’
  - 'Valid for domestic cash transactions' bit is set to 1b
  - 'Valid for internationl cash transactions' bit is set to 1b
  - 'Valid for domestic goods’ bit is set to 1b.
  - 'Valid for international goods’ bit is set to 1b
  - 'Valid for domestic services’ bit is set to 1b
  - 'Valid for international services’ bit is set to 1b
  - 'Valid at ATMs' bit is set to 1b.
  - 'Valid at terminals other than ATMs' bit is set to 1b
  - 'Domestic cashback allowed' bit is set to 1b.
  - 'International cashback allowed' bit is set to 1b.

Action: NA

Procedure: Application in LT is selected and transaction is processed with LT.

Pass Criteria:

- The terminal shall process the transaction until completion, by requesting a TC.
- TVR byte 2, bit 5 = '0' (ie Requested service allowed for card product) received at 1st GENERATE AC.
- Transaction Type shall indicate a cash transaction.

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2CJ.047.00 Transaction type is a Cash transaction, and Issuer Country Code differs from Terminal Country Code

Test No: **2CJ.047.00** - *Revision* 43j - *Number of sub cases:* 0

Objective: To ensure that if the Terminal Country Code does not match Issuer Country Code, Transaction Type indicates a cash transaction, the AUC is present in the card, and the 'Valid for international cash transactions' bit is not set to 1b in the AUC, the terminal sets the 'Requested service not allowed for card product' bit in the TVR to 1b.

Reference EMV 4.3: Book 3 - Section 10.4.2 - Application Usage Control

Terminal Configuration: [Cash] supported

Terminal Configuration (New format): [Cash]

Conditions:

- Transaction is a cash transaction.
- Issuer Country Code does not match Terminal Country Code.
- AUC is present in LT with value ‘BFC0’
  - 'Valid for domestic cash transactions' bit is set to 1b
  - 'Valid for internationl cash transactions' bit is set to 0b
  - 'Valid for domestic goods’ bit is set to 1b.
  - 'Valid for international goods’ bit is set to 1b
  - 'Valid for domestic services’ bit is set to 1b
  - 'Valid for international services’ bit is set to 1b
  - 'Valid at ATMs' bit is set to 1b.
  - 'Valid at terminals other than ATMs' bit is set to 1b
  - 'Domestic cashback allowed' bit is set to 1b.
  - 'International cashback allowed' bit is set to 1b.

Action: NA

Procedure: Application in LT is selected and transaction is processed with LT.

Pass Criteria:

- The terminal shall process the transaction until completion, by requesting a TC.
- TVR byte 2, bit 5 = '1' (ie Requested service not allowed for card product) received at 1st GENERATE AC.
- Transaction Type shall indicate a cash transaction.

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2CJ.048.00 Transaction type is a Cash transaction, and Issuer Country Code differs from Terminal Country Code (implied)

Test No: **2CJ.048.00** - *Revision* 43j - *Number of sub cases:* 0

Objective: To ensure that if the Terminal Country Code does not match Issuer Country Code, Transaction Type indicates a cash transaction, the AUC is present in the card, and the 'Valid for international cash transactions' bit is set to 1b in the AUC, the terminal does not set the 'Requested service not allowed for card product' bit in the TVR to 1b.

Reference EMV 4.3: Book 3 - Section 10.4.2 - Application Usage Control

Terminal Configuration: [Cash] supported

Terminal Configuration (New format): [Cash]

Conditions:

- Transaction is a cash transaction.
- Issuer Country Code does not match Terminal Country Code.
- AUC is present in LT with value ‘FFC0’
  - 'Valid for domestic cash transactions' bit is set to 1b
  - 'Valid for internationl cash transactions' bit is set to 1b
  - 'Valid for domestic goods’ bit is set to 1b.
  - 'Valid for international goods’ bit is set to 1b
  - 'Valid for domestic services’ bit is set to 1b
  - 'Valid for international services’ bit is set to 1b
  - 'Valid at ATMs' bit is set to 1b.
  - 'Valid at terminals other than ATMs' bit is set to 1b
  - 'Domestic cashback allowed' bit is set to 1b.
  - 'International cashback allowed' bit is set to 1b.

Action: NA

Procedure: Application in LT is selected and transaction is processed with LT.

Pass Criteria:

- The terminal shall process the transaction until completion, by requesting a TC.
- TVR byte 2, bit 5 = '0' (ie Requested service allowed for card product) received at 1st GENERATE AC.
- Transaction Type shall indicate a cash transaction.

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2CJ.049.00 Transaction type is a Purchase of goods and services, and Issuer Country Code = Terminal Country Code

Test No: **2CJ.049.00** - *Revision* 43j - *Number of sub cases:* 0

Objective: To ensure that if the Terminal Country Code matches Issuer Country Code, Transaction Type indicates a purchase of goods and services, the AUC is present in the card, and the 'Valid for domestic goods' bit is not set to 1b in the AUC, the terminal sets the 'Requested service not allowed for card product' bit in the TVR to 1b.

Reference EMV 4.3: Book 3 - Section 10.4.2 - Application Usage Control

Terminal Configuration: [Goods] supported OR [Services] supported

Terminal Configuration (New format): [Goods] OR [Services]

Conditions:

- Transaction is a purchase of goods and services.
- Issuer Country Code matches Terminal Country Code.
- AUC is present in LT with value ‘D7C0’
  - 'Valid for domestic cash transactions' bit is set to 1b
  - 'Valid for internationl cash transactions' bit is set to 1b
  - 'Valid for domestic goods’ bit is set to 0b.
  - 'Valid for international goods’ bit is set to 1b
  - 'Valid for domestic services’ bit is set to 0b
  - 'Valid for international services’ bit is set to 1b
  - 'Valid at ATMs' bit is set to 1b.
  - 'Valid at terminals other than ATMs' bit is set to 1b
  - 'Domestic cashback allowed' bit is set to 1b.
  - 'International cashback allowed' bit is set to 1b.

Action: NA

Procedure: Application in LT is selected and transaction is processed with LT.

Pass Criteria:

- The terminal shall process the transaction until completion, by requesting a TC.
- TVR byte 2, bit 5 = '1' (ie Requested service not allowed for card product) received at 1st GENERATE AC.
- Transaction Type shall indicate a purchase of goods and services.

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2CJ.050.00 Transaction type is a Purchase of goods and Services, and Issuer Country Code = Terminal Country Code (implied)

Test No: **2CJ.050.00** - *Revision* 43j - *Number of sub cases:* 0

Objective: To ensure that if the Terminal Country Code matches Issuer Country Code, Transaction Type indicates a purchase of goods and services, the AUC is present in the card, and the 'Valid for domestic goods' bit is set to 1b in the AUC, the terminal does not set the 'Requested service not allowed for card product' bit in the TVR to 1b.

Reference EMV 4.3: Book 3 - Section 10.4.2 - Application Usage Control

Terminal Configuration: [Goods] supported OR [Services] supported

Terminal Configuration (New format): [Goods] OR [Services]

Conditions:

- Transaction is a purchase of goods or services (Transaction type is ‘00’).
- Issuer Country Code matches Terminal Country Code.
- AUC is present in LT with value ‘F7C0’
  - 'Valid for domestic cash transactions' bit is set to 1b
  - 'Valid for internationl cash transactions' bit is set to 1b
  - 'Valid for domestic goods’ bit is set to 1b.
  - 'Valid for international goods’ bit is set to 1b
  - 'Valid for domestic services’ bit is set to 0b
  - 'Valid for international services’ bit is set to 1b
  - 'Valid at ATMs' bit is set to 1b.
  - 'Valid at terminals other than ATMs' bit is set to 1b
  - 'Domestic cashback allowed' bit is set to 1b.
  - 'International cashback allowed' bit is set to 1b.

Action: NA

Procedure: Application in LT is selected and transaction is processed with LT.

Pass Criteria:

- The terminal shall process the transaction until completion, by requesting a TC.
- TVR byte 2, bit 5 = '0' (ie Requested service allowed for card product) received at 1st GENERATE AC.
- Transaction Type shall indicate a purchase of goods and services.

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2CJ.051.00 Transaction type is a Purchase of goods and Services, and Issuer Country Code differs from Terminal Country Code

Test No: **2CJ.051.00** - *Revision* 43j - *Number of sub cases:* 0

Objective: To ensure that if the Terminal Country Code does not match Issuer Country Code, Transaction Type indicates a purchase of goods and services, the AUC is present in the card, and the 'Valid for international goods' bit is not set to 1b in the AUC, the terminal sets the 'Requested service not allowed for card product' bit in the TVR to 1b.

Reference EMV 4.3: Book 3 - Section 10.4.2 - Application Usage Control

Terminal Configuration: [Goods] supported OR [Services] supported

Terminal Configuration (New format): [Goods] OR [Services]

Conditions:

- Transaction is a purchase of goods or services (Transaction type is ‘00’).
- Issuer Country Code does not match Terminal Country Code.
- AUC is present in LT with value ‘EBC0’
  - 'Valid for domestic cash transactions' bit is set to 1b
  - 'Valid for internationl cash transactions' bit is set to 1b
  - 'Valid for domestic goods’ bit is set to 1b.
  - 'Valid for international goods’ bit is set to 0b
  - 'Valid for domestic services’ bit is set to 1b
  - 'Valid for international services’ bit is set to 0b
  - 'Valid at ATMs' bit is set to 1b.
  - 'Valid at terminals other than ATMs' bit is set to 1b
  - 'Domestic cashback allowed' bit is set to 1b.
  - 'International cashback allowed' bit is set to 1b.

Action: NA

Procedure: Application in LT is selected and transaction is processed with LT.

Pass Criteria:

- The terminal shall process the transaction until completion, by requesting a TC.
- TVR byte 2, bit 5 = '1' (ie Requested service not allowed for card product) received at 1st GENERATE AC.
- Transaction Type shall indicate a purchase of goods and services.

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2CJ.052.00 Transaction type is a Purchase of goods and services, and Issuer Country Code differs from Terminal Country Code (implied)

Test No: **2CJ.052.00** - *Revision* 43j - *Number of sub cases:* 0

Objective: To ensure that if the Terminal Country Code does not match Issuer Country Code, Transaction Type indicates a purchase of goods and services, the AUC is present in the card, and the 'Valid for international goods' bit is set to 1b in the AUC, the terminal does not set the 'Requested service not allowed for card product' bit in the TVR to 1b.

Reference EMV 4.3: Book 3 - Section 10.4.2 - Application Usage Control

Terminal Configuration: [Goods] supported OR [Services] supported

Terminal Configuration (New format): [Goods] OR [Services]

Conditions:

- The CDOL1 of LT requests the Trasnaction Type
- Transaction is a purchase of goods or services (Transaction type is ‘00’).
- Issuer Country Code does not match Terminal Country Code.
- AUC is present in LT with value ‘FBC0’
  - 'Valid for domestic cash transactions' bit is set to 1b
  - 'Valid for internationl cash transactions' bit is set to 1b
  - 'Valid for domestic goods’ bit is set to 1b.
  - 'Valid for international goods’ bit is set to 1b
  - 'Valid for domestic services’ bit is set to 1b
  - 'Valid for international services’ bit is set to 0b
  - 'Valid at ATMs' bit is set to 1b.
  - 'Valid at terminals other than ATMs' bit is set to 1b
  - 'Domestic cashback allowed' bit is set to 1b.
  - 'International cashback allowed' bit is set to 1b.

Action: NA

Procedure: Application in LT is selected and transaction is processed with LT.

Pass Criteria:

- The terminal shall process the transaction until completion, by requesting a TC.
- TVR byte 2, bit 5 = '0' (ie Requested service allowed for card product) received at 1st GENERATE AC.
- Transaction Type shall indicate a purchase of goods and services.

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2CJ.053.00 Transaction type is a Purchase of goods and services, and Issuer Country Code = Terminal Country Code

Test No: **2CJ.053.00** - *Revision* 43j - *Number of sub cases:* 0

Objective: To ensure that if the Terminal Country Code matches Issuer Country Code, Transaction Type indicates a purchase of goods and services, the AUC is present in the card, and the 'Valid for domestic services' bit is not set to 1b in the AUC, the terminal sets the 'Requested service not allowed for card product' bit in the TVR to 1b.

Reference EMV 4.3: Book 3 - Section 10.4.2 - Application Usage Control

Terminal Configuration: [Goods] supported OR [Services] supported

Terminal Configuration (New format): [Goods] OR [Services]

Conditions:

- Transaction is a purchase of goods or services (Transaction type is ‘00’).
- Issuer Country Code matches Terminal Country Code.
- AUC is present in LT with value ‘D7C0’
  - 'Valid for domestic cash transactions' bit is set to 1b
  - 'Valid for internationl cash transactions' bit is set to 1b
  - 'Valid for domestic goods’ bit is set to 0b.
  - 'Valid for international goods’ bit is set to 1b
  - 'Valid for domestic services’ bit is set to 0b
  - 'Valid for international services’ bit is set to 1b
  - 'Valid at ATMs' bit is set to 1b.
  - 'Valid at terminals other than ATMs' bit is set to 1b
  - 'Domestic cashback allowed' bit is set to 1b.
  - 'International cashback allowed' bit is set to 1b.

Action: NA

Procedure: Application in LT is selected and transaction is processed with LT.

Pass Criteria:

- The terminal shall process the transaction until completion, by requesting a TC.
- TVR byte 2, bit 5 = '1' (ie Requested service not allowed for card product) received at 1st GENERATE AC.
- Transaction Type shall indicate a purchase of goods and services.

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2CJ.054.00 Transaction type is a Purchase of goods and services, and Issuer Country Code = Terminal Country Code (implied)

Test No: **2CJ.054.00** - *Revision* 43j - *Number of sub cases:* 0

Objective: To ensure that if the Terminal Country Code matches Issuer Country Code, Transaction Type indicates a purchase of goods and services, the AUC is present in the card, and the 'Valid for domestic services' bit is set to 1b in the AUC, the terminal does not set the 'Requested service not allowed for card product' bit in the TVR to 1b.

Reference EMV 4.3: Book 3 - Section 10.4.2 - Application Usage Control

Terminal Configuration: [Goods] supported OR [Services] supported

Terminal Configuration (New format): [Goods] OR [Services]

Conditions:

- Transaction is a purchase of goods or services (Transaction type is ‘00’).
- Issuer Country Code matches Terminal Country Code.
- AUC is present in LT with value FFC0’
  - 'Valid for domestic cash transactions' bit is set to 1b
  - 'Valid for internationl cash transactions' bit is set to 1b
  - 'Valid for domestic goods’ bit is set to 1b.
  - 'Valid for international goods’ bit is set to 1b
  - 'Valid for domestic services’ bit is set to 1b
  - 'Valid for international services’ bit is set to 1b
  - 'Valid at ATMs' bit is set to 1b.
  - 'Valid at terminals other than ATMs' bit is set to 1b
  - 'Domestic cashback allowed' bit is set to 1b.
  - 'International cashback allowed' bit is set to 1b.

Action: NA

Procedure: Application in LT is selected and transaction is processed with LT.

Pass Criteria:

- The terminal shall process the transaction until completion, by requesting a TC.
- TVR byte 2, bit 5 = '0' (ie Requested service allowed for card product) received at 1st GENERATE AC.
- Transaction Type shall indicate a purchase of goods and services.

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2CJ.056.00 Transaction type is a Purchase of goods and services, and Issuer Country Code differs from Terminal Country Code (implied)

Test No: **2CJ.056.00** - *Revision* 43j - *Number of sub cases:* 0

Objective: To ensure that if the Terminal Country Code does not match Issuer Country Code, Transaction Type indicates a purchase of goods and services, the AUC is present in the card, and the 'Valid for international services' bit is set to 1b in the AUC, the terminal does not set the 'Requested service not allowed for card product' bit in the TVR to 1b.

Reference EMV 4.3: Book 3 - Section 10.4.2 - Application Usage Control

Terminal Configuration: [Goods] supported OR [Services] supported

Terminal Configuration (New format): [Goods] OR [Services]

Conditions:

- Transaction is a purchase of goods or services (Transaction type is ‘00’).
- Issuer Country Code does not match Terminal Country Code.
- AUC is present in LT with value ‘EFC0’
  - 'Valid for domestic cash transactions' bit is set to 1b
  - 'Valid for internationl cash transactions' bit is set to 1b
  - 'Valid for domestic goods’ bit is set to 1b.
  - 'Valid for international goods’ bit is set to 0b
  - 'Valid for domestic services’ bit is set to 1b
  - 'Valid for international services’ bit is set to 1b
  - 'Valid at ATMs' bit is set to 1b.
  - 'Valid at terminals other than ATMs' bit is set to 1b
  - 'Domestic cashback allowed' bit is set to 1b.
  - 'International cashback allowed' bit is set to 1b.

Action: NA

Procedure: Application in LT is selected and transaction is processed with LT.

Pass Criteria:

- The terminal shall process the transaction until completion, by requesting a TC.
- TVR byte 2, bit 5 = '0' (ie Requested service allowed for card product) received at 1st GENERATE AC.
- Transaction Type shall indicate a purchase of goods and services.

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2CJ.057.00 Transaction has a Cashback amount, and Issuer Country Code = Terminal Country Code

Test No: **2CJ.057.00** - *Revision* 43j - *Number of sub cases:* 0

Objective: To ensure that if the Terminal Country Code matches Issuer Country Code,Transaction has a Cashback amount, the AUC is present in the card, and the 'Domestic Cashback allowed' bit is not set to 1b in the AUC, the terminal sets the 'Requested service not allowed for card product' bit in the TVR to 1b.

Reference EMV 4.3: Book 3 - Section 10.4.2 - Application Usage Control

Terminal Configuration: [CashBack] supported

Terminal Configuration (New format): [Cashback]

Conditions:

- Transaction has a cashback amount.
- Issuer Country Code matches Terminal Country Code.
- AUC is present in LT with value ‘FF40’
  - 'Valid for domestic cash transactions' bit is set to 1b
  - 'Valid for internationl cash transactions' bit is set to 1b
  - 'Valid for domestic goods’ bit is set to 1b.
  - 'Valid for international goods’ bit is set to 1b
  - 'Valid for domestic services’ bit is set to 1b
  - 'Valid for international services’ bit is set to 1b
  - 'Valid at ATMs' bit is set to 1b.
  - 'Valid at terminals other than ATMs' bit is set to 1b
  - 'Domestic cashback allowed' bit is set to 0b.
  - 'International cashback allowed' bit is set to 1b.

Action: NA

Procedure: Application in LT is selected and transaction is processed with LT.

Pass Criteria:

- The terminal shall process the transaction until completion, by requesting a TC.
- TVR byte 2, bit 5 = '1' (ie Requested service not allowed for card product) received at 1st GENERATE AC.

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2CJ.058.00 Transaction has a Cashback amount, and Issuer Country Code = Terminal Country Code (implied)

Test No: **2CJ.058.00** - *Revision* 43j - *Number of sub cases:* 0

Objective: To ensure that if the Terminal Country Code matches Issuer Country Code, Transaction has a Cashback amount, the AUC is present in the card, and the 'Domestic cashback allowed' bit is set to 1b in the AUC, the terminal does not set the 'Requested service not allowed for card product' bit in the TVR to 1b.

Reference EMV 4.3: Book 3 - Section 10.4.2 - Application Usage Control

Terminal Configuration: [CashBack] supported

Terminal Configuration (New format): [Cashback]

Conditions:

- Transaction has a cashback amount.
- Issuer Country Code matches Terminal Country Code.
- AUC is present in LT with value ‘FFC0’
  - 'Valid for domestic cash transactions' bit is set to 1b
  - 'Valid for internationl cash transactions' bit is set to 1b
  - 'Valid for domestic goods’ bit is set to 1b.
  - 'Valid for international goods’ bit is set to 1b
  - 'Valid for domestic services’ bit is set to 1b
  - 'Valid for international services’ bit is set to 1b
  - 'Valid at ATMs' bit is set to 1b.
  - 'Valid at terminals other than ATMs' bit is set to 1b
  - 'Domestic cashback allowed' bit is set to 1b.
  - 'International cashback allowed' bit is set to 1b.

Action: NA

Procedure: Application in LT is selected and transaction is processed with LT.

Pass Criteria:

- The terminal shall process the transaction until completion, by requesting a TC.
- TVR byte 2, bit 5 = '0' (ie Requested service allowed for card product) received at 1st GENERATE AC.

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2CJ.059.00 Transaction type is a Cashback amount, and Issuer Country Code differs from Terminal Country Code

Test No: **2CJ.059.00** - *Revision* 43j - *Number of sub cases:* 0

Objective: To ensure that if the Terminal Country Code does not match Issuer Country Code, Transaction has a Cashback amount, the AUC is present in the card, and the 'International cashback allowed' bit is not set to 1b in the AUC, the terminal sets the 'Requested service not allowed for card product' bit in the TVR to 1b.

Reference EMV 4.3: Book 3 - Section 10.4.2 - Application Usage Control

Terminal Configuration: [CashBack] supported

Terminal Configuration (New format): [Cashback]

Conditions:

- Transaction has a cashback amount.
- Issuer Country Code does not match Terminal Country Code.
- AUC is present in LT with value ‘FF80’
  - 'Valid for domestic cash transactions' bit is set to 1b
  - 'Valid for internationl cash transactions' bit is set to 1b
  - 'Valid for domestic goods’ bit is set to 1b.
  - 'Valid for international goods’ bit is set to 1b
  - 'Valid for domestic services’ bit is set to 1b
  - 'Valid for international services’ bit is set to 1b
  - 'Valid at ATMs' bit is set to 1b.
  - 'Valid at terminals other than ATMs' bit is set to 1b
  - 'Domestic cashback allowed' bit is set to 1b.
  - 'International cashback allowed' bit is set to 0b.

Action: NA

Procedure: Application in LT is selected and transaction is processed with LT.

Pass Criteria:

- The terminal shall process the transaction until completion, by requesting a TC.
- TVR byte 2, bit 5 = '1' (ie Requested service not allowed for card product) received at 1st GENERATE AC.

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2CJ.060.00 Transaction type is a Cashback amount, and Issuer Country Code differs from Terminal Country Code (implied)

Test No: **2CJ.060.00** - *Revision* 43j - *Number of sub cases:* 0

Objective: To ensure that if the Terminal Country Code does not match Issuer Country Code, Transaction has a Cashback amount, the AUC is present in the card, and the 'International cashback allowed' bit is set to 1b in the AUC, the terminal does not set the 'Requested service not allowed for card product' bit in the TVR to 1b.

Reference EMV 4.3: Book 3 - Section 10.4.2 - Application Usage Control

Terminal Configuration: [CashBack] supported

Terminal Configuration (New format): [Cashback]

Conditions:

- Transaction has a cashback amount.
- Issuer Country Code does not match Terminal Country Code.
- AUC is present in LT with value ‘FFC0’
  - 'Valid for domestic cash transactions' bit is set to 1b
  - 'Valid for internationl cash transactions' bit is set to 1b
  - 'Valid for domestic goods’ bit is set to 1b.
  - 'Valid for international goods’ bit is set to 1b
  - 'Valid for domestic services’ bit is set to 1b
  - 'Valid for international services’ bit is set to 1b
  - 'Valid at ATMs' bit is set to 1b.
  - 'Valid at terminals other than ATMs' bit is set to 1b
  - 'Domestic cashback allowed' bit is set to 1b.
  - 'International cashback allowed' bit is set to 1b.

Action: NA

Procedure: Application in LT is selected and transaction is processed with LT.

Pass Criteria:

- The terminal shall process the transaction until completion, by requesting a TC.
- TVR byte 2, bit 5 = '0' (ie Requested service allowed for card product) received at 1st GENERATE AC.

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2CJ.062.00 Application Usage Control is present in the ICC but not Issuer Country code (implied)

Test No: **2CJ.062.00** - *Revision* 43j - *Number of sub cases:* 0

Objective: To ensure that if the Application Usage control is present in the ICC but not Issuer Country Code, the Terminal skips the second set of tests described in Book 3 Section 6.4.3.

Reference EMV 4.3: Book 3 - Section 10.4.2 - Application Usage Control

Terminal Configuration: NA

Terminal Configuration (New format): [NA]

Conditions:

- AUC is present in LT with value ‘FFC0’
  - 'Valid for domestic cash transactions' bit is set to 1b
  - 'Valid for internationl cash transactions' bit is set to 1b
  - 'Valid for domestic goods’ bit is set to 1b.
  - 'Valid for international goods’ bit is set to 1b
  - 'Valid for domestic services’ bit is set to 1b
  - 'Valid for international services’ bit is set to 1b
  - 'Valid at ATMs' bit is set to 1b.
  - 'Valid at terminals other than ATMs' bit is set to 1b
  - 'Domestic cashback allowed' bit is set to 1b.
  - 'International cashback allowed' bit is set to 1b.
- Issuer Country Code is not present in the LT.

Action: NA

Procedure: Application in LT is selected and transaction is processed with LT.

Pass Criteria:

- The terminal shall process the transaction until completion,by requesting a TC.
- TVR byte 2, bit 5 = '0' (ie Requested service allowed for card product) received at 1st GENERATE AC.

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2CK.005.01 CDOL requests a TC Hash Value and neither TDOL nor default TDOL is present (2)

Test No: **2CK.005.01** - *Revision* 43j - *Number of sub cases:* 2

Objective: To ensure that if TC Hash Value is requested in CDOL, TDOL is not provided by the card and and Terminal supports default TDOL and default TDOL is not configured, the terminal is able to calculate TC Hash value using a TDOL with no data object.

Reference EMV 4.3: Book 3 - Section 9.2.2 - Transaction Certificate Data

Terminal Configuration:

Terminal Configuration (New format): [Default TDOL] AND [Default TDOL NotConfigurable]

Conditions:

- Default TDOL is not configured/loaded in the terminal
- LT does not contain TDOL
- Case 01: CDOL1 requests TC Hash value
- Case 02: CDOL1 does not request TC Hash value

Action: NA

Procedure: Application in LT is selected and transaction is processed with LT.

Pass Criteria:

- Case 01: The terminal shall return a GENERATE AC command to the LT with data field containing a calculated TC Hash value with no input data
- TVR byte 5, bit 8 = '0' (ie Default TDOL not used) received at 1st GENERATE AC.

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2CM.063.00 Cashback Amount is Transmitted in the Amount, Other Data Object

Test No: **2CM.063.00** - *Revision* 43j - *Number of sub cases:* 0

Objective: To ensure that the terminal uses Amount Other field for Cash Back Amount, if cash back is supported

Reference EMV 4.3: Book 4 - Section 6.5.1 - Amount Entry and Management

Terminal Configuration: [CashBack] supported

Terminal Configuration (New format): [Cashback]

Conditions:

- AUC is present in LT with value ‘FFC0’
  - 'Valid for domestic cash transactions' bit is set to 1b
  - 'Valid for internationl cash transactions' bit is set to 1b
  - 'Valid for domestic goods’ bit is set to 1b.
  - 'Valid for international goods’ bit is set to 1b
  - 'Valid for domestic services’ bit is set to 1b
  - 'Valid for international services’ bit is set to 1b
  - 'Valid at ATMs' bit is set to 1b.
  - 'Valid at terminals other than ATMs' bit is set to 1b
  - 'Domestic cashback allowed' bit is set to 1b.
  - 'International cashback allowed' bit is set to 1b.
- CDOL1 requests Amount Other (in Numeric or Binary format)

Action: Cardholder asks for Cash back

Procedure: Application in LT is selected and transaction is processed with LT.

Pass Criteria:

- The terminal shall process the transaction until completion, by requesting a TC.
- Amount Other shall contain value entered for Cashback

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2CM.064.00 Purchase and Cashback Transmitted in Amount Authorised

Test No: **2CM.064.00** - *Revision* 43j - *Number of sub cases:* 0

Objective: To ensure that the terminal include Purchase Amount and Cashback Amount in Amount Authorized

Reference EMV 4.3: Book 4 - Section 6.5.1 - Amount Entry and Management

Terminal Configuration: [CashBack] supported

Terminal Configuration (New format): [Cashback]

Conditions:

- AUC is present in LT with value ‘FFC0’
  - 'Valid for domestic cash transactions' bit is set to 1b
  - 'Valid for internationl cash transactions' bit is set to 1b
  - 'Valid for domestic goods’ bit is set to 1b.
  - 'Valid for international goods’ bit is set to 1b
  - 'Valid for domestic services’ bit is set to 1b
  - 'Valid for international services’ bit is set to 1b
  - 'Valid at ATMs' bit is set to 1b.
  - 'Valid at terminals other than ATMs' bit is set to 1b
  - 'Domestic cashback allowed' bit is set to 1b.
  - 'International cashback allowed' bit is set to 1b.
- CDOL1 requests Amount Authorized (in Numeric or Binary format), and Amount Other (in Numeric or Binary format)

Action: Cardholder asks for Cash back

Procedure: Application in LT is selected and transaction is processed with LT.

Pass Criteria:

- The terminal shall process the transaction until completion, by requesting a TC.
- Amount Authorized shall be the sum of Cashback Amount and Purchase Amount

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2CM.067.00 Other Amount Is Expressed With Implicit Decimal Point

Test No: **2CM.067.00** - *Revision* 43j - *Number of sub cases:* 0

Objective: To ensure that the terminal stores Amount Other with Implicit Decimal Point

Reference EMV 4.3: Book 4 - Section 6.5.1 - Amount Entry and Management

Terminal Configuration: [CashBack] AND ([Display] OR [Receipt]) supported

Terminal Configuration (New format): [Cashback] AND ([Display] OR [Receipt])

Conditions:

- AUC is present in LT with value ‘FFC0’
  - 'Valid for domestic cash transactions' bit is set to 1b
  - 'Valid for internationl cash transactions' bit is set to 1b
  - 'Valid for domestic goods’ bit is set to 1b.
  - 'Valid for international goods’ bit is set to 1b
  - 'Valid for domestic services’ bit is set to 1b
  - 'Valid for international services’ bit is set to 1b
  - 'Valid at ATMs' bit is set to 1b.
  - 'Valid at terminals other than ATMs' bit is set to 1b
  - 'Domestic cashback allowed' bit is set to 1b.
  - 'International cashback allowed' bit is set to 1b.
- CDOL1 requests Amount Other
- Case 01: Purchase Amount has decimal values (if applicable depending on currency code) using currency code 1 declared in the ICS.
- …
- Case n: Purchase Amount has decimal values (if applicable depending on currency code) using currency code n declared in the ICS.

Action: Cardholder asks for Cash back

Procedure: Application in LT is selected and transaction is processed with LT.

Pass Criteria:

- The terminal shall process the transaction until completion, by requesting a TC.
- Amount Other shall be expressed with implicit decimal point for the currency used (if applicable for the tested currency code)
- The corresponding transaction amount with shall be indicated to the cardholder preferably by means of the terminal display, alternatively by printing on the receipt.

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2CM.083.00 Terminal Maintains Transaction Sequence Counter

Test No: **2CM.083.00** - *Revision* 43j - *Number of sub cases:* 0

Objective: To ensure that the terminal maintains a Transaction Sequence Counter

Reference EMV 4.3: Book 4 - Section 6.5.5 - Transaction Sequence Counter

Terminal Configuration:

Terminal Configuration (New format): [Offline Only]

Conditions:

- 10 Transactions are performed
- CDOL1 requests Transaction Sequence Counter

Action: NA

Procedure: Application in LT is selected and transaction is processed with LT.

Pass Criteria:

- The terminal shall process the transaction until completion, by requesting a TC for each transaction
- Transaction Sequence Counter shall be incremented by one in numeric format at each transaction

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2CM.083.01 Terminal Maintains Transaction Sequence Counter (2)

Test No: **2CM.083.01** - *Revision* 43j - *Number of sub cases:* 0

Objective: To ensure that the terminal maintains a Transaction Sequence Counter

Reference EMV 4.3: Book 4 - Section 6.5.5 - Transaction Sequence Counter

Terminal Configuration:

Terminal Configuration (New format): [Online Only]

Conditions:

- 10 Transactions are performed
- CDOL1 requests Transaction Sequence Counter

Action: NA

Procedure: Application in LT is selected and transaction is processed with LT.

Pass Criteria:

- The terminal shall process the transaction until completion, by requesting a TC at 2nd Generate AC for each transaction
- Transaction Sequence Counter shall be incremented by one in numeric format at each transaction

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2CM.083.02 (3) Terminal Maintains Transaction Sequence Counter

Test No: **2CM.083.02** - *Revision* 43j - *Number of sub cases:* 0

Objective: To ensure that the terminal maintains a Transaction Sequence Counter

Reference EMV 4.3: Book 4 - Section 6.5.5 - Transaction Sequence Counter

Terminal Configuration:

Terminal Configuration (New format): [Offline/Online Capable]

Conditions:

- 10 Transactions are performed
- 5 transactions are performed online and 5 transactions are performed offline in the same test.
- CDOL1 requests Transaction Sequence Counter

Action: NA

Procedure: Application in LT is selected and transaction is processed with LT.

Pass Criteria:

- The terminal shall process the transaction until completion, by requesting a TC at 1st Generate AC or at 2nd Generate AC for each transaction
- Transaction Sequence Counter shall be incremented by one in numeric format at each transaction

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2CM.083.03 Terminal reset Transaction Sequence Counter when limit is reached

Test No: **2CM.083.03** - *Revision* 43j - *Number of sub cases:* 0

Objective: To ensure that when the Transaction Sequence Counter reaches its maximum value, the terminal reset it to one

Reference EMV 4.3: Book 4 - Section 6.5.5 - Transaction Sequence Counter

Terminal Configuration:

Terminal Configuration (New format): [TSCvalue]

Conditions:

- Test is applicable only if the terminal is able to personalise the Transaction Sequence Counter to any value
- Transaction Sequence Counter is set to max value minus 5
- 10 Transactions are performed
- CDOL1 requests Transaction Sequence Counter

Action: NA

Procedure: Application in LT is selected and transaction is processed with LT.

Pass Criteria:

- The terminal shall process the transaction until completion, by requesting a TC for each transaction
- Transaction Sequence Counter shall be set back to 1 at transaction 6
- Transaction Sequence Counter shall be incremented by one in numeric format at each transaction

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